

Learning to do repeated web tasks faster and more cheaply

An agent runs the same workflow again and again with new inputs. What it learns is kept outside the model, so that each later task takes less time and money without lowering the quality of the result.

I · The problem (pages 1–4) · II · How far existing work has got (pages 5–10) · III · Our plan follows after discussion.

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The problem: the same workflow with new inputs, in less time and for less money at the same success rate

Part 1’s cost per success, with Part 2’s set-up spread over the n tasks it serves · time is reported beside money, never added to it

For tasks that repeat one workflow with new inputs, learning changes the harness around a fixed model. It pays off when the learned design is faster and cheaper per task, its set-up included, at a success rate no lower than a level fixed in advance.

FORMULA	WHAT IT SAYS	SOURCE
$v(m, p) = \frac{C_m(p)}{R_m(p)}, \quad T_{\text{success}}(m, p) = \frac{\mathbb{E}[T_{\text{attempt}}]}{R_m(p)}$	dollars and seconds per success of design m (model + harness) on task p	Part 1, page 2; (Erol et al., 2026)
$v_n(m', p) = \frac{C_{\text{setup}}}{n} + \frac{C_{m'}(p)}{R_{m'}(p)}$	m' : the same model, its harness changed by learning; the learning bill C_{setup} is spread over the n tasks it serves	Part 2, App. B8; fixed and variable cost (Kapoor et al., 2025), §3; one set-up over n runs (Cai et al., 2024)
$\min (T_{\text{success}}(m', p), v_n(m', p)) \text{ s. t. } R_{m'}(p) \geq R_0$	Part 1’s goal with the set-up included; success measured on tasks not used for learning	Part 1, page 2; R_0 self-defined
$n^* = \frac{C_{\text{setup}}}{v(m, p) - v(m', p)}$	break-even: the number of tasks after which learning has paid for itself	calc. from row 2; the same comparison in (Kapoor et al., 2025), §3.2

- C_{setup} — exploring, proposing and checking changes, and keeping them up to date, each counted once
- T_{setup} — the offline learning time, reported beside T_{success} and not spread over tasks (self-defined)

LEARNING IS CHARGED · C_{setup}

A saving per task counts only once the learning behind it is paid back

- 27 s vs 87 s and \$0.05 vs \$0.40 per task without exploration and warm-up; with exploration, break-even after 39–101 tasks, warm-up still excluded — ActionEngine, WebArena (655 tasks), Claude Opus 4.6, vs Claude Code (Zhong et al., 2026)

SCOPE

First version: the model weights stay fixed; prompts, memory, skills and tools, and control code may change

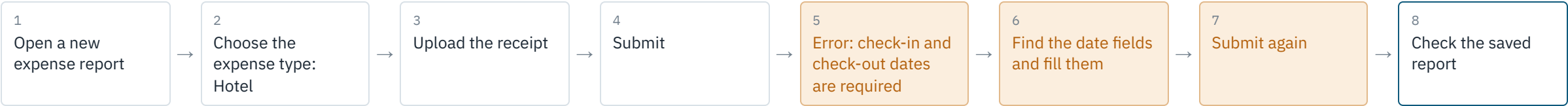
- Our experimental scope, not a definition of acceleration: serving, model choice, caching and parallel calls are other routes

Where one expense report loses time

term: $C_m'(p)$ – the part of a run that is repeated discovery or rework

Some steps of a run are needed to do the task; others rediscover the form or redo work after an error. Only the second kind can be learned away, and which steps are which is not known before the agent has run.

ONE RUN, AS THE AGENT EXPERIENCES IT



WITH A LEARNED RULE 1 → 2 → 3 → fill the dates → 4 → 8 steps 5–7 (orange) are the rework a rule could remove; each costs a model call, page time and tokens

FOUR WORDS USED ON EVERY PAGE

Task	Environment	Run record	Learning
A goal, its input data and a starting state, with a check of the result that the agent does not control.	The website’s interface, its rules and the actions it permits; it stays the same while records change.	What one execution observed and did, the feedback it got, its outcome, its time and its cost.	An update that outlives a task and changes something outside the model: a prompt, a memory, a skill or tool, or control code.

Illustrative: the environment is not yet chosen and the form rule is invented; no numbers on this page.

Five difficulties, each tied to a term of the formula

difficulties from our problem framing · each card names the term it sits in

A faster agent is easy to claim and hard to prove: the bottleneck is the whole run, what can be removed is not known in advance, a failure does not name its fix, learned material must transfer and stay cheap to read, and proving a saving costs money.

1 · THE WHOLE RUN · $C_{m'}(p)$

Fewer actions can still mean a longer task, so judge by end-to-end time and money

- Actions 10.7 → 10.2 per task, yet 215 → 395 s per task — WebCoach, 643 live WebVoyager tasks, Skywork-38B with a Qwen3-8B coach; the time includes the coach’s inference (Liu et al., 2025)

2 · NEEDED OR REMOVABLE · $C_{m'}(p)$

What can be skipped is known only after a run has succeeded once, and that run is paid for

- Replay cut the median task time 315.7 → 127.5 s, but only on the 159 tasks whose first pass had succeeded and been stored; building each took a median of about 572k tokens and 4.5 min — EchoPath, OSWorld-Verified (Zhao et al., 2026)

3 · FROM FAILURE TO FIX · C_{setup}

Seeing a failure does not say what to change; finding and checking a fix costs model calls

- Completion +6.3 to +18.4 points after automatic repairs, mean of three runs, GPT-5 mini on four benchmarks; the repairs on AppWorld used 37.2 million tokens offline — HarnessFix (Chen et al., 2026)

4 · TRANSFER AND READING · $C_{m'}(p)$

Learned material has to help on new tasks and be cheap to read

- With token budgets roughly matched, the plain agent without skills or memory did best: 44.78% success at 73.6K tokens per task — WebArena, four domains, Gemini 3 Flash, three runs (Hajimiri et al., 2026)

5 · PROVING THE SAVING · C_{setup}

Showing that a change saves money costs money itself

- \$1.91 → \$1.70 and 130.2 → 114.5 s per task on four benchmarks, for a whole run of about \$7,000: $n^* \approx 33,000$ tasks at that saving per attempt (calc.) — SICA (Robeyns et al., 2025)

Acceleration and self-improvement overlap in 16 of 178 agent methods

our coding of the calibrated corpus (383 works; 198 study agents, 178 of them propose a method) · calc.

Of 178 agent methods, 16 both aim at running time and learn across tasks. 88 learn without aiming at time, and 30 aim at time without learning.

	LEARNS OR IMPROVES ACROSS TASKS	DOES NOT
running time is a main goal	16	30
running time is not a main goal	88	44

“Learns or improves”: reuses experience, explores the environment, or keeps rewriting prompts, skills, memory or harness code; training the model alone does not count. A work without a time goal may still cut cost or steps.

NOT ONLY SUCCESS

Self-improving agents do not all optimise success alone

- SICA and SpeedRunner also target cost or usage (Robeyns et al., 2025; Huang et al., 2026)

TWO DENOMINATORS

This page counts 178 agent methods; pages 5–10 count the 102 works checked one by one for the two directions

- The two counts come from different corpora and are not added or compared

Direction 1 · Learning from the agent’s own runs: 93 works in four classes

$C_{m'}(p) \downarrow$ is the aim · $C_{\text{setup}} \uparrow$ is the price · $R_{m'}(p) \geq R_0$ must hold

Of 93 works that change what sits outside the model, 21 measured the seconds or dollars of running tasks, 19 what learning cost, and 28 tested on unseen tasks.

PER TASK $v_n(m', p) =$

$\frac{C_{\text{setup}}}{n}$

↑

+

$\frac{C_{m'}(p)}{R_{m'}(p)}$

↓

s. t.

$R_{m'}(p) \geq R_0 =$

targeted

in exchange

unchanged

CLASS	WORKS	WHAT CHANGES	REPRESENTATIVE WORK: ONE NUMBER AND ITS CONDITIONS	TIME OR MONEY OF RUNS YES · PARTLY	LEARNING COST YES · PARTLY	UNSEEN TASKS YES · PARTLY
Prompts and context A02 ↗	15	the prompt, instructions or context read on the next task	ACE: offline adaptation lifts AppWorld from 42.4 to 59.4 (with labels), and takes 9,517 s against GEPA’s 53,898 s — DeepSeek-V3.1 (Q. Zhang et al., 2026)	1 · 2	5 · 5	7 · 6
Experience memory A03 ↗	26	retrievable experience: successes, failures, workflows	AWM: 35.5% vs 23.5% success on WebArena (the baseline also reads HTML), 5.9 vs 7.9 steps against an accessibility-tree BrowserGym — GPT-4-0613 (Wang et al., 2025a)	5 · 15	5 · 4	11 · 14
Skills and tools A04 ↗	34	callable functions, scripts, skill documents	AXIS: 29.9 s vs 59.5 s and \$0.2 vs \$0.4 per task, 84% vs 52% success, against UFO on 50 Microsoft Word tasks; learning cost not reported (Lu et al., 2025)	9 · 14	6 · 11	8 · 23
Whole harness A05 ↗	30	the outer program: tools, control loop, the improver itself	SICA: \$1.91 → \$1.70 and 130.2 → 114.5 s per task, for a whole run of about \$7,000; the benchmark that picked the agent also scored it (Robeyns et al., 2025)	8 · 4	7 · 16	5 · 23
All, each work once	93			21 · 28	19 · 33	28 · 57

One row per work in our literature table; a work in two classes counts in both. Yes / partly: our initial judgment from the table cells (calc.), defined in Appendix A1.

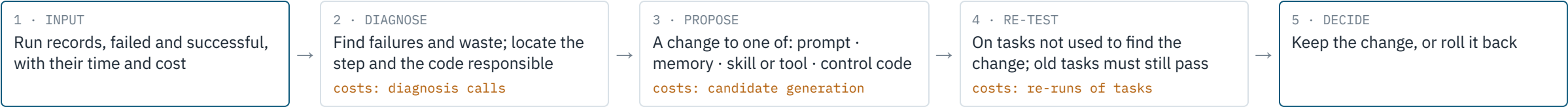
How a run record becomes a change, and what the change costs

direction 1 · $C_m'(p)$ ↓ after the change · C_{setup} ↑ at three points: diagnosis, candidates, re-tests

The loop reads run records, finds failures and waste, proposes a change to a prompt, a memory, a skill or the code, re-tests it, and keeps it or rolls it back. Every arrow but the first costs model calls.

PER TASK $v_n(m', p) = \frac{C_{\text{setup}}}{n} \uparrow + \frac{C_{m'}(p)}{R_{m'}(p)} \downarrow$ s. t. $R_{m'}(p) \geq R_0 =$

targeted in exchange unchanged



IT WORKS

Changes can be found and made automatically, with gains on the final tasks

- Highest success in five of six settings and 76.0–91.8% fewer model calls at deployment — Growing Harness, BrowseComp-Plus and WebArena-Verified, three models, 50 final tasks per setting, three runs (L. Li et al., 2026)

THE BILL IS PARTIAL · C_{setup}

Savings at deployment are usually reported without the learning that produced them

- 97.4K vs 112.6K tokens and 11.25 vs 14.55 turns per AppWorld task once the memory is frozen; the memory manager’s own calls are not included — Metis (Dai et al., 2026)

Direction 2 · Learning the environment: 29 works keep four kinds of knowledge

$C_{m'}(p) \downarrow$ is the aim · $C_{\text{setup}} \uparrow$ is the exploring · a kept rule must still hold when the site changes

Of 29 works that keep knowledge about the environment, 14 measured the seconds or dollars of running tasks, 5 what exploring cost, and 8 tested on unseen tasks.

PER TASK $v_n(m', p) =$

$\frac{C_{\text{setup}}}{n}$ $\uparrow$

+

$\frac{C_{m'}(p)}{R_{m'}(p)}$ $\downarrow$

 s. t.

$R_{m'}(p) \geq R_0 =$

targeted

in exchange

unchanged

CLASS	WORKS	WHAT CHANGES	REPRESENTATIVE WORK: ONE NUMBER AND ITS CONDITIONS	TIME OR MONEY OF RUNS YES · PARTLY	LEARNING COST YES · PARTLY	UNSEEN TASKS YES · PARTLY
Predicting what an action does A06 ↗	3	what the page will look like after an action	WMA: 140.3 s vs 748.3 s and \$0.4 vs \$2.7 per instruction against tree search, 16.6% vs 19.2% success — WebArena, GPT-4o (Chae et al., 2025)	1 · 1	1 · 2	1 · 2
Descriptions, facts and preconditions A07 ↗	13	tool parameters and errors; facts and traps of the site	DRAFT: correct tool paths 88.00 vs 71.00 for ReAct on RestBench TMDB, GPT-4o; time and cost not reported (Qu et al., 2025)	6 · 4	2 · 3	3 · 8
Page and place structure A08 ↗	7	a map of pages, actions and the paths between them	MobileGPT: −62.5% latency and −68.8% cost on repeated tasks with human-repaired paths; exploring took 10–15 min per app, \$10.78 in all (Lee et al., 2024)	5 · 2	1 · 3	1 · 6
Exploring and practising A09 ↗	9	self-set practice tasks, turned into tools	WALT: 50.1% and 52.9% success on WebArena and VisualWebArena; exploring and checking cost \$1.67 per tool, repaid after about 14 uses (Prabhu et al., 2026)	3 · 2	1 · 3	4 · 4
All, each work once	29			14 · 8	5 · 10	8 · 18

One row per work in our literature table; a work in two classes counts in both. Yes / partly: our initial judgment from the table cells (calc.), defined in Appendix A1. Three environments that check the final state, such as AppWorld (Trivedi et al., 2024), learn nothing; Appendix A6.

Exploring a site, keeping a checked rule, and using it on the next task

direction 2 · the exploring is C_setup · the rule removes rework from C_m'(p) · it must be re-checked when the site changes

The agent probes the form, writes down what it found together with where it applies and the evidence for it, and reads that note on the next task, so the error-and-retry loop of page 2 does not happen.

1 · EXPLORE THE FORM

Change the expense type from Meal to Hotel. Two new required fields appear: check-in and check-out date. Submitting without them returns an error.

2 · KEEP A CHECKED RULE

site-guide.md

scope: new expense form

IF type = Hotel → check-in and check-out dates required

evidence: 2 probes, 1 error message

re-check: when the form changes

3 · USE IT ON THE NEXT TASK

New hotel expense + the live page + the saved note → fill every required field → submit once → check the saved report.

WITHOUT THE NOTE



WITH THE NOTE



illustrative paths, not measured

REAL CALLS CORRECT THE DESCRIPTION

Trying an action and reading its result can fix what a description gets wrong

- DRAFT found by calling a tool that it needs a valid person_id, and wrote that condition back into the tool’s documentation (Qu et al., 2025)

ONE OBSERVATION IS NOT A RULE

A rule written from one observation can be wrong, and a local check can miss it

- DRAFT once generalised from a single error message; SkillWeaver’s local checks let broken functions through (Qu et al., 2025; Zheng et al., 2025)

Across both directions, what learning costs is rarely counted

C_setup rarely itemised · $R_{m'}(p) \geq R_0$ hard to prove · fewer steps $\neq$ less time or money

The learning bill is rarely itemised, proving that a change is correct is the weak link, and a saving measured as fewer steps, or in one setting, does not guarantee less time or money elsewhere.

LEARNING COST · C_{setup}

Few works count both what learning costs and what it saves

- 5 of 102 works measured both the seconds or dollars of running tasks and a learning cost: GPTSwarm, MobileGPT, OpenSkill, EchoPath and Unbrowse (initial judgment, calc.) (Zhuge et al., 2024; Lee et al., 2024; Yan et al., 2026; Zhao et al., 2026; Tham et al., 2026)

CORRECTNESS · $R_{m'}(p) \geq R_0$

Showing that a change fixed the target without breaking what worked is the open problem

- In AgentDevel’s ablations, the highest mean score and the fewest regressions on old tasks may have to be traded against each other (Zhang, 2026)

FEWER STEPS · $C_{m'}(p)$

Fewer steps is not the same as less time or money

- Steps 9.7 $\rightarrow$ 8.3 on WebArena, while total tokens rose 50,847.4 $\rightarrow$ 53,054.5 in another table whose base set-up is not stated — ReasoningBank (Ouyang et al., 2026)

TRANSFER · $C_{m'}(p)$

A saving in one setting need not carry over

- Median cost per task \$0.143 vs \$0.144 after moving to SkillsBench, no fall — ClawTrace, OpenClaw with gpt-5.4 (Yuan et al., 2026)

The closest works save time or money on repeated tasks; none counts the whole loop

`v_n(m',p)` · what each closest work measured, and what it left out

In the works we checked, no single result covers all of these at once: any task stream, no training, automatic location of failures and waste, protection of old abilities, every learning and maintenance cost counted, and a net saving over time.

WORK	WHAT IT LEARNS	WHAT IT SHOWS: ONE NUMBER AND ITS CONDITIONS	WHAT IS LEFT OUT
ActionEngine (Zhong et al., 2026)	a state-machine map of the site, with templates	91.2% success (73.1% without the warmed-up map); 27 s vs 87 s and \$0.05 vs \$0.40 per task against Claude Code — WebArena, 655 tasks, Claude Opus 4.6	exploration and warm-up are outside the per-task numbers; break-even after 39–101 tasks still leaves warm-up out
WALT (Prabhu et al., 2026)	tools written from exploring each site	\$1.67 per tool to explore and check it, repaid after about 14 uses	repairing tools after deployment is left to future work
SpeedRunner (Huang et al., 2026)	programmatic skills	BabyAI usage about one eighth, success from about 67% to near-perfect — gpt-5.4-mini, 200 online tasks, 30 held out, three seeds	output tokens and dollars disagree in one figure; total learning cost [TBD: appendix A.3, to be re-read]
HarnessFix (Chen et al., 2026)	repairs to the outer program from failed runs	completion +6.3 to +18.4 points, mean of three runs	time and cost of running tasks not reported
Growing Harness (L. Li et al., 2026)	code changes from function-level failures	inference cost at deployment −74.4% to −98.6%	evaluator and offline optimiser not counted; break-even not measured
Metis (Dai et al., 2026)	text and code memory, frozen before the final tasks	97.4K vs 112.6K tokens per AppWorld task	memory manager not counted; apps never seen before not tested

References (1/5)

Agashe et al., 2025. Saaket Agashe; Jiuzhou Han; Shuyu Gan; et al. 2025. Agent S: An Open Agentic Framework that Uses Computers Like a Human. International Conference on Learning Representations 2025 (ICLR 2025), main conference, poster (decision 'Accept (Poster)'; OpenReview LVRgt4nLv). [formal citation](#). Version read: [HTML](#) (v1, 2024-10-10; first version 2024-10-10).

Agrawal et al., 2026. Lakshya A Agrawal; Shangyin Tan; Dilara Soylu; et al. 2026. GEPA: Reflective Prompt Evolution Can Outperform Reinforcement Learning. ICLR 2026, main conference; decision 'Accept (Oral)' (ICLR.cc/2026/Conference). [formal citation](#). Version read: [HTML](#) (v2, 2026-02-14; first version 2025-07-25).

Amayuelas et al., 2026. Alfonso Amayuelas; Firas Laakom; Piotr Piękos; et al. 2026. Planning to Explore: Curiosity-Driven Planning for LLM Test Generation. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v1, 2026-04-06; first version 2026-04-06).

Azam et al., 2025. Ruhana Azam; Aditya Vempaty; Ashish Jagmohan. 2025. Reflection-Based Memory For Web navigation Agents. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v1, 2025-06-02; first version 2025-06-02).

Bogdanov et al., 2026. Igor Bogdanov; Chung-Horng Lung; Thomas Kunz; et al. 2026. FORGE: Self-Evolving Agent Memory With No Weight Updates via Population Broadcast. Proceedings of the ACM Conference on AI and Agentic Systems (ACM CAIS '26, San Jose CA, May 26-29, 2026), research paper, pp. 292-310; DOI 10.1145/3786335.3813155. [formal citation](#). Version read: [HTML](#) (v1, 2026-05-15; first version 2026-05-15).

Cai et al., 2024. Tianle Cai; Xuezhi Wang; Tengyu Ma; et al. 2024. Large Language Models as Tool Makers. International Conference on Learning Representations 2024 (ICLR 2024), Conference (main); decision 'Accept (poster)', Poster Session 8, #280 (OpenReview qV83K9d5WB). [formal citation](#). Version read: [HTML](#) (v2, 2024-03-11; first version 2023-05-26).

Cai et al., 2026b. Qianshu Cai; Yonggang Zhang; Xianzhang Jia; et al. 2026. MOSS: Self-Evolution through Source-Level Rewriting in Autonomous Agent Systems. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v2, 2026-05-23; first version 2026-05-21).

Chae et al., 2025. Hyunjoo Chae; Namyoun Kim; Kai Ong; et al. 2025. Web Agents with World Models: Learning and Leveraging Environment Dynamics in Web Navigation. International Conference on Learning Representations 2025 (ICLR 2025), Conference. [formal citation](#). Version read: [HTML](#) (v2, 2025-03-29; first version 2024-10-17).

Chen et al., 2024. Yongchao Chen; Jacob Arkin; Yilun Hao; et al. 2024. PRompt Optimization in Multi-Step Tasks (PROMST): Integrating Human Feedback and Heuristic-based Sampling. EMNLP 2024, main conference: Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, pages 3859-3920, Miami, Florida, USA, November 2024; Anthology ID 2024.emnlp-main.226; DOI 10.18653/v1/2024.emnlp-main.226. [formal citation](#). Version read: [HTML](#) (v4, 2024-10-03; first version 2024-02-13).

Chen et al., 2026. Mengzhuo Chen; Junjie Wang; Zhe Liu; et al. 2026. From Failed Trajectories to Reliable LLM Agents: Diagnosing and Repairing Harness Flaws. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v2, 2026-07-02; first version 2026-06-04).

Cheng et al., 2024. Ching-An Cheng; Allen Nie; Adith Swaminathan. 2024. Trace is the Next AutoDiff: Generative Optimization with Rich Feedback, Execution Traces, and LLMs. Advances in Neural Information Processing Systems 37 (NeurIPS 2024), Main Conference Track, poster; DOI 10.52202/079017-2287. [formal citation](#). Version read: [HTML](#) (v2, 2024-11-01; first version 2024-06-23).

Cheng et al., 2026. Zihao Cheng; Zeming Liu; Yingyu Shan; et al. 2026. Mem^2Evolve: Towards Self-Evolving Agents via Co-Evolutionary Capability Expansion and Experience Distillation. ACL 2026: Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), main conference, pages 20784-20831, San Diego, California, United States, July 2026; Anthology ID 2026.acl-long.952; DOI 10.18653/v1/2026.acl-long.952. [formal citation](#). Version read: [HTML](#) (v1, 2026-04-13; first version 2026-04-13).

Dai et al., 2026. Zijie Dai; Siuhin He; Hui Li; et al. 2026. Metis: Bridging Text and Code Memory for Self-Evolving Agents. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v1, 2026-06-23; first version 2026-06-23).

Erol et al., 2026. Erol, M. H., El, B., Suzgun, M., Yuksekgonul, M., & Zou, J. (2026). Cost-of-Pass: An economic framework for evaluating language models. *The Fourteenth International Conference on Learning Representations (ICLR 2026)*. arXiv:2504.13359v2. Stanford University.

Esakkiraja et al., 2026. Esakkivel Esakkiraja; Denis Akhiyarov; Vikas Yadav; et al. 2026. StarHarness: Evolving Harnesses with Stratified Search for Enterprise Environments. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v1, 2026-08-25; first version 2026-08-25).

G. Wang et al., 2024. Guanzhi Wang; Yuqi Xie; Yunfan Jiang; et al. 2024. Voyager: An Open-Ended Embodied Agent with Large Language Models. Transactions on Machine Learning Research (TMLR), March 2024, journal paper (ISSN 2835-8856); certifications J2C (Journal to Conference) with event certification iclr.cc/ICLR/2025/Journal_Track. [formal citation](#). Version read: [HTML](#) (v2, 2023-10-19; first version 2023-05-25).

G. Zhang et al., 2025. Guibin Zhang; Muxin Fu; Guancheng Wan; et al. 2025. G-Memory: Tracing Hierarchical Memory for Multi-Agent Systems. Advances in Neural Information Processing Systems 38 (NeurIPS 2025), Main Conference Track, Spotlight Poster; DOI 10.52202/085713-0439. [formal citation](#). Version read: [HTML](#) (v2, 2025-06-16; first version 2025-06-09).

Guo et al., 2025. Yifu Guo; Jiaye Lin; Huacan Wang; et al. 2025. SE-Agent: Self-Evolution Trajectory Optimization in Multi-Step Reasoning with LLM-Based Agents. Advances in Neural Information Processing Systems 38 (NeurIPS 2025), Main Conference Track, poster (decision 'Accept (poster)'; OpenReview isATAFP71B); DOI 10.52202/085713-3877. [formal citation](#). Version read: [PDF](#) (v6, 2025-11-03; first version 2025-08-04).

H. Liu et al., 2026. Haozhe Liu; Tian Ye; Sensen Gao; et al. 2026. SoL-Pi: Recursively Scaling Auto-Research Loops for Efficient Agent Harness. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v1, 2026-09-17; first version 2026-09-17).

H. Zhang et al., 2026. Hanrong Zhang; Shicheng Fan; Henry Peng Zou; et al. 2026. CoEvoSkills: Self-Evolving Agent Skills via Co-Evolutionary Verification. COLM 2026 (Conference on Language Modeling), main conference paper; OpenReview venue 'COLM 2026', venueid colmweb.org/COLM/2026/Conference, publication date 2026-07-08; official COLM 2026 Accepted Papers page lists it as a poster (Imperial Ballroom, Poster Session 3, #38). [formal citation](#). Version read: [HTML](#) (v3, 2026-08-10; first version 2026-04-02).

Hajimiri et al., 2026. Sina Hajimiri; Masih Aminbeidokhti; Jose Dolz; et al. 2026. Are Online Skill and Memory Modules Always Worth Their Tokens? A Budget-Constrained Study of Web Agents. EMNLP 2026 (The 2026 Conference on Empirical Methods in Natural Language Processing), main conference, paper ID 4950-MAIN, poster (Session 4, Poster Session B, 2026-10-25). [formal citation](#). Version read: [HTML](#) (v2, 2026-08-30; first version 2026-06-12).

Hao et al., 2026. Guangya Hao; Yunbo Long; Zhuokai Zhao. 2026. Self-Evolving Multi-Agent Systems via Decentralized Memory (DecentMem). arXiv preprint. [formal citation](#). Version read: [HTML](#) (v1, 2026-05-21; first version 2026-05-21).

References (2/5)

Hattami et al., 2026. Amine El Hattami; Nicolas Chapados; Christopher Pal. 2026. SKILL.nb: Selective Formalization and Gated Execution for Durable Agent Workflows. FAGEN@ICML 2026 Poster (ICML 2026 workshop, venueid ICML.cc/2026/Workshop/FAGEN; not the ICML main conference). [formal citation](#). Version read: [HTML](#) (v1, 2026-06-06; first version 2026-06-06).

Hu et al., 2025. Shengran Hu; Cong Lu; Jeff Clune. 2025. Automated Design of Agentic Systems. International Conference on Learning Representations 2025 (ICLR 2025), Conference track (ICLR proceedings page); OpenReview venue field: "ICLR 2025 Poster". [formal citation](#). Version read: [HTML](#) (v2, 2025-03-02; first version 2024-08-15).

Huang et al., 2026. Zixi Huang; Xiheng Wang; Andrew Wang; et al. 2026. Better, Faster, Stronger: Programmatic Skill Learning Best Reduces Agent Cost. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v1, 2026-08-11; first version 2026-08-11).

Jacob et al., 2026. Alex Jacob; Andrej Jovanović; William F. Shen; et al. 2026. The Red Queen Gödel Machine: Co-Evolving Agents and Their Evaluators. arXiv preprint. [formal citation](#). Version read: [PDF](#) (v2, 2026-06-29; first version 2026-06-24).

J. Liu et al., 2026. Jiaqi Liu; Xinyu Ye; Peng Xia; et al. 2026. EvolveMem: Self-Evolving Memory Architecture via AutoResearch for LLM Agents. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v1, 2026-05-13; first version 2026-05-13).

Jiang et al., 2025. Wenjia Jiang; Yangyang Zhuang; Chenxi Song; et al. 2025. AppAgentX: Evolving GUI Agents as Proficient Smartphone Users. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v3, 2025-04-15; first version 2025-03-04).

Jin et al., 2025. Ruofan Jin; Zaixi Zhang; Mengdi Wang; et al. 2025. STELLA: Self-Evolving LLM Agent for Biomedical Research. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v1, 2025-07-01; first version 2025-07-01).

Jin et al., 2026. Yuxin Jin; Siyuan Zhang; Hanchen Wang; et al. 2026. EXG: Self-Evolving Agents with Experience Graphs. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v1, 2026-05-18; first version 2026-05-18).

Kapoor et al., 2025. Kapoor, Sayash; Stroebel, Benedikt; Siegel, Zachary S.; et al. 2025. AI Agents That Matter. Transactions on Machine Learning Research. Version read: [HTML](#) (v1, 2024-07-01; first version 2024-07-01).

Karten et al., 2026. Seth Karten; Joel Zhang; Tersoo Upaa Jr; et al. 2026. Continual Harness: Online Adaptation for Self-Improving Foundation Agents. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v1, 2026-05-11; first version 2026-05-11).

Khattab et al., 2024. Omar Khattab; Arnav Singhvi; Paridhi Maheshwari; et al. 2024. DSPy: Compiling Declarative Language Model Calls into State-of-the-Art Pipelines. International Conference on Learning Representations 2024 (ICLR 2024), main conference, spotlight poster (decision 'Accept (spotlight)'; OpenReview sY5N0zY5Od), published under the title 'DSPy: Compiling Declarative Language Model Calls into State-of-the-Art Pipelines'. [formal citation](#). Version read: [HTML](#) (v1, 2023-10-05; first version 2023-10-05).

L. Li et al., 2026. Laizhen Li; Jiarui Li; Juanjuan Zhao; et al. 2026. Grow the Harness, Not the Context: From Strategy-Free Scaffolds to Reusable Specialist Agents. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v2, 2026-09-24; first version 2026-09-22).

L. Liu et al., 2026. Lihao Liu; Peng Tang; Kunwar Yashraj Singh; et al. 2026. ESPO: Error-Structured Prompt Optimization via Diagnose, Diversify, and Stabilize. EMNLP 2026, Findings track (paper ID 3300-FIND; in-person poster, Session 2 Poster Session A, 2026-10-25, Budapest); Findings of ACL: EMNLP 2026 not yet in the ACL Anthology. [formal citation](#). Version read: [HTML](#) (v1, 2026-09-03; first version 2026-09-03).

Lee et al., 2024. Sunjae Lee; Junyoung Choi; Jungjae Lee; et al. 2024. MobileGPT: Augmenting LLM with Human-like App Memory for Mobile Task Automation. Proceedings of the 30th Annual International Conference on Mobile Computing and Networking (ACM MobiCom 2024), research paper, pp. 1119-1133. [formal citation](#). Version read: [HTML](#) (v3, 2024-10-16; first version 2023-12-04).

Liang et al., 2026. Jiaqing Liang; Jinyi Han; Weijia Li; et al. 2026. GenericAgent: A Token-Efficient Self-Evolving LLM Agent via Contextual Information Density Maximization (V1.0). arXiv preprint. [formal citation](#). Version read: [HTML](#) (v1, 2026-04-18; first version 2026-04-18).

Liao et al., 2026. Junwei Liao; Haoting Shi; Ruiwen Zhou; et al. 2026. MemQ: Integrating Q-Learning into Self-Evolving Memory Agents over Provenance DAGs. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v3, 2026-05-14; first version 2026-05-08).

Lindenbauer et al., 2025. Tobias Lindenbauer; Georg Groh; Hinrich Schütze. 2025. From Knowledge to Noise: CTIM-Rover and the Pitfalls of Episodic Memory in Software Engineering Agents. Proceedings of the 1st Workshop for Research on Agent Language Models (REALM 2025), Vienna, July 2025, pages 411-427; Anthology ID 2025.realm-1.30; DOI 10.18653/v1/2025.realm-1.30. This is an ACL 2025 workshop, not the main conference or Findings. [formal citation](#). Version read: [HTML](#) (v1, 2025-05-29; first version 2025-05-29).

Liu et al., 2025. Genglin Liu; Shijie Geng; Sha Li; et al. 2025. WebCoach: Self-Evolving Web Agents with Cross-Session Memory Guidance. ICLR 2026 Workshop on Memory for LLM-Based Agentic Systems (MemAgents), workshop paper in Poster and Discussion Session 1. [formal citation](#). Version read: [HTML](#) (v2, 2026-07-22; first version 2025-11-17).

Lou et al., 2026. Zhanzhi Lou; Hui Chen; Yibo Li; et al. 2026. Learning to Learn-at-Test-Time: Language Agents with Learnable Adaptation Policies. ICML 2026 Workshop AIWILD, under the v1 title 'Learning to Learn-at-Test-Time: Language Agents with Learnable Adaptation Policies'. [formal citation](#). Version read: [HTML](#) (v4, 2026-09-28; first version 2026-04-01).

Lu et al., 2025. Lu, J.; Zhang, Z.; Yang, F.; et al. 2025. AXIS: Efficient human-agent-computer interaction with API-first LLM-based agents. Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). [formal citation](#). Version read: [PDF](#).

Lyu et al., 2025. Bohan Lyu; Xin Cong; Heyang Yu; et al. 2025. Enhancing Open-Domain Task-Solving Capability of LLMs via Autonomous Tool Integration from GitHub. ACL 2025 (63rd Annual Meeting of the Association for Computational Linguistics), main conference, Volume 1: Long Papers, pages 17257-17277, Vienna, July 2025; Anthology ID 2025.acl-long.845; DOI 10.18653/v1/2025.acl-long.845. [formal citation](#). Version read: [HTML](#) (v3, 2025-06-10; first version 2023-12-28).

Ma et al., 2026. Zihan Ma; Zhikai Zhao; Chuanbo Hua; et al. 2026. JudgeFlow: Agentic Workflow Optimization via Block Judge. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v2, 2026-02-02; first version 2026-01-12).

References (3/5)

Madaan et al., 2023. Aman Madaan; Niket Tandon; Prakhar Gupta; et al. 2023. Self-Refine: Iterative Refinement with Self-Feedback. Advances in Neural Information Processing Systems 36 (NeurIPS 2023), Main Conference Track; DOI 10.52202/075280-2019. [formal citation](#). Version read: [HTML](#) (v2, 2023-05-25; first version 2023-03-30).

Nguyen et al., 2026a. Michael Nguyen; Quoc Nguyen; Paul Vuong. 2026. Recursive Self-Evolving Agents via Held-Out Selection. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v1, 2026-06-17; first version 2026-06-17).

Opsahl-Ong et al., 2024. Krista Opsahl-Ong; Michael J Ryan; Josh Purtell; et al. 2024. Optimizing Instructions and Demonstrations for Multi-Stage Language Model Programs. EMNLP 2024: Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (main conference), pages 9340-9366, Miami, Florida, USA, November 2024; Anthology ID 2024.emnlp-main.525; DOI 10.18653/v1/2024.emnlp-main.525. [formal citation](#). Version read: [HTML](#) (v2, 2024-10-06; first version 2024-06-17).

Ou et al., 2025. Yixin Ou; Wangchunshu Zhou; Shengwei Ding; et al. 2025. Symbolic learning enables self-evolving agents. AI Open, Volume 6, pp. 314-322 (2025), journal article (Elsevier B.V. on behalf of KeAi; open access, CC BY 4.0). [formal citation](#). Version read: [HTML](#) (v1, 2024-06-26; first version 2024-06-26).

Ouyang et al., 2026. Siru Ouyang; Jun Yan; I-Hung Hsu; et al. 2026. ReasoningBank: Scaling Agent Self-Evolving with Reasoning Memory. ICLR 2026, main conference, poster. [formal citation](#). Version read: [HTML](#) (v2, 2026-03-16; first version 2025-09-29).

Park et al., 2025. Junyeong Park; Junmo Cho; Sungjin Ahn. 2025. MrSteve: Instruction-Following Agents in Minecraft with What-Where-When Memory. International Conference on Learning Representations 2025 (ICLR 2025), Conference (main), poster. [formal citation](#). Version read: [HTML](#) (v5, 2025-04-11; first version 2024-11-11).

Prabhu et al., 2026. Viraj Prabhu; Yutong Dai; Matthew Fernandez; et al. 2026. WALT: Web Agents that Learn Tools. International Conference on Learning Representations 2026 (ICLR 2026), Conference track (proceedings pp. 55589-55607); OpenReview venue field: "ICLR 2026 Poster". [formal citation](#). Version read: [PDF](#) (v1, 2025-10-01; first version 2025-10-01 (Wed, 1 Oct 2025 23:41:47 UTC)).

Pryzant et al., 2023. Reid Pryzant; Dan Iter; Jerry Li; et al. 2023. Automatic Prompt Optimization with "Gradient Descent" and Beam Search. Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP 2023), main conference, pages 7957-7968; Anthology ID 2023.emnlp-main.494; DOI 10.18653/v1/2023.emnlp-main.494. [formal citation](#). Version read: [HTML](#) (v2, 2023-10-19; first version 2023-05-04).

Q. Zhang et al., 2026. Qizheng Zhang; Changran Hu; Shubhangi Upasani; et al. 2026. Agentic Context Engineering: Evolving Contexts for Self-Improving Language Models. ICLR 2026, main conference, poster (decision 'Accept (Poster)'). [formal citation](#). Version read: [PDF](#) (v3, 2026-03-29; first version 2025-10-06).

Qian et al., 2025. Hongjin Qian; Zheng Liu. 2025. MetaAgent: Toward Self-Evolving Agent via Tool Meta-Learning. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v2, 2025-09-01; first version 2025-08-01).

Qiu et al., 2025a. Jiahao Qiu; Xinzhe Juan; Yimin Wang; et al. 2025. AgentDistill: Training-Free Agent Distillation with Generalizable MCP Boxes. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v1, 2025-06-17; first version 2025-06-17).

Qiu et al., 2025b. Jiahao Qiu; Xuan Qi; Hongru Wang; et al. 2025. Alita-G: Self-Evolving Generative Agent for Agent Generation. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v1, 2025-10-27; first version 2025-10-27).

Qu et al., 2025. Changle Qu; Sunhao Dai; Xiaochi Wei; et al. 2025. From Exploration to Mastery: Enabling LLMs to Master Tools via Self-Driven Interactions. ICLR 2025 (International Conference on Learning Representations 2025), main conference (proceedings 'Conference' track); decision Accept (Oral), Oral Session 4B, also Poster Session 3. [formal citation](#). Version read: [HTML](#) (v2, 2025-02-26; first version 2024-10-10).

Robeyns et al., 2025. Maxime Robeyns; Martin Szummer; Laurence Aitchison. 2025. A Self-Improving Coding Agent. ICLR 2025 Workshop 'Self-Improving Foundation Models Without Human Supervision', Oral in Workshop (workshop paper, not ICLR main conference). [formal citation](#). Version read: [HTML](#) (v2, 2025-05-16; first version 2025-04-21).

S. Wu et al., 2024. Shirley Wu; Shiyu Zhao; Qian Huang; et al. 2024. AvaTaR: Optimizing LLM Agents for Tool Usage via Contrastive Reasoning. Advances in Neural Information Processing Systems 37 (NeurIPS 2024), Main Conference Track, poster (decision 'Accept (poster)'); DOI 10.52202/079017-0817. [formal citation](#). Version read: [HTML](#) (v3, 2024-10-31; first version 2024-06-17).

S. Zhang et al., 2026. Shengtao Zhang; Jiaqian Wang; Ruiwen Zhou; et al. 2026. MemRL: Self-Evolving Agents via Runtime Reinforcement Learning on Episodic Memory. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v2, 2026-02-12; first version 2026-01-06).

Shao et al., 2026. Shuai Shao; Kangning Zhang; Qingyao Li; et al. 2026. Harness-R1: Learning to Edit Executable Runtime Harnesses from Agent Failure Trajectories. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v1, 2026-08-03; first version 2026-08-03).

Shinn et al., 2023. Noah Shinn; Federico Cassano; Ashwin Gopinath; et al. 2023. Reflexion: Language Agents with Verbal Reinforcement Learning. Advances in Neural Information Processing Systems 36 (NeurIPS 2023), Main Conference Track. [formal citation](#). Version read: [HTML](#) (v4, 2023-10-10; first version 2023-03-20).

Sun et al., 2026. Zeyi Sun; Ziyu Liu; Yuhang Zang; et al. 2026. SEAgent: Self-Evolving Computer Use Agent with Autonomous Learning from Experience. ICML 2026: Proceedings of the 43rd International Conference on Machine Learning, PMLR 306:116867-116893 (volume published 2026-09-29), main conference, Poster (ICML virtual site: Thu 9 Jul 2026, Hall A #117). [formal citation](#). Version read: [HTML](#) (v2, 2025-08-12; first version 2025-08-06).

Suresh et al., 2026. Susheel Suresh; Hazel Mak; Sahil Bhatnagar; et al. 2026. Grounding Agent Memory: Environment-Probing Curation for Enterprise Agents. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v1, 2026-09-10; first version 2026-09-10).

Suzgun et al., 2026. Mirac Suzgun; Mert Yuksekgonul; Federico Bianchi; et al. 2026. Dynamic Cheatsheet: Test-Time Learning with Adaptive Memory. EACL 2026: Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers), pages 7080-7106, Rabat, Morocco, March 2026; main conference long paper; Anthology ID 2026.eacl-long.333; DOI 10.18653/v1/2026.eacl-long.333. [formal citation](#). Version read: [HTML](#) (v1, 2025-04-10; first version 2025-04-10).

Tham et al., 2026. Lewis Tham; Nicholas Mac Gregor Garcia; Jungpil Hahn. 2026. Internal APIs Are All You Need: Shadow APIs, Shared Discovery, and the Case Against Browser-First Agent Architectures. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v1, 2026-04-01; first version 2026-04-01).

References (4/5)

Trivedi et al., 2024. Harsh Trivedi; Tushar Khot; Mareike Hartmann; et al. 2024. AppWorld: A Controllable World of Apps and People for Benchmarking Interactive Coding Agents. ACL 2024: Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), main conference, pages 16022-16076, Bangkok, Thailand, August 2024; Anthology ID 2024.acl-long.850; DOI 10.18653/v1/2024.acl-long.850. [formal citation](#). Version read: [HTML](#) (v1, 2024-07-26; first version 2024-07-26).

W. Wang et al., 2024. Wenyi Wang; Hisham A. Alyahya; Dylan R. Ashley; et al. 2024. How to Correctly do Semantic Backpropagation on Language-based Agentic Systems. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v1, 2024-12-04; first version 2024-12-04).

W. Zhang et al., 2025. Wentao Zhang; Liang Zeng; Yuzhen Xiao; et al. 2025. AgentOrchestra: Orchestrating Multi-Agent Intelligence with the Tool-Environment-Agent(TEA) Protocol. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v6, 2026-05-28; first version 2025-06-14).

Wang et al., 2025a. Zora Zhiruo Wang; Jiayuan Mao; Daniel Fried; et al. 2025. Agent Workflow Memory. Proceedings of the 42nd International Conference on Machine Learning (ICML 2025), PMLR 267:63897-63911, 2025 (the PMLR page names no track; conference proceedings volume). [formal citation](#). Version read: [HTML](#) (v1, 2024-09-11; first version 2024-09-11).

Wang et al., 2025b. Zora Zhiruo Wang; Apurva Gandhi; Graham Neubig; et al. 2025. Inducing Programmatic Skills for Agentic Tasks. COLM 2025 (Conference on Language Modeling), accepted paper, main conference (listed on the official "COLM 2025: Accepted Papers" page without a Spotlight badge). [formal citation](#). Version read: [PDF](#); [HTML](#) (v2, 2025-08-29; first version 2025-04-09).

Wang et al., 2026. Wenyi Wang; Piotr Piękos; Li Nanbo; et al. 2026. Huxley-Gödel Machine: Human-Level Coding Agent Development by an Approximation of the Optimal Self-Improving Machine. ICLR 2026 (International Conference on Learning Representations 2026), main conference (proceedings track 'Conference'), Oral; official ICLR 2026 paper data give decision 'Accept (Oral)', Oral Session 1C Code generation and ML Systems, plus Poster Session 2 Pavilion 3 (OpenReview T0EiEuhOOL). [formal citation](#). Version read: [HTML](#) (v3, 2025-06-29; first version 2025-10-24).

Wen et al., 2025. Hao Wen; Shizuo Tian; Borislav Pavlov; et al. 2025. AutoDroid-V2: Boosting SLM-based GUI Agents via Code Generation. Proceedings of the 23rd Annual International Conference on Mobile Systems, Applications and Services (ACM MobiSys '25), June 23-27, 2025, Anaheim, CA, USA; main research-paper track (listed on the official MobiSys 2025 Accepted Papers page, with a Best Artifact Award); pp. 223-235. [formal citation](#). Version read: [HTML](#) (v3, 2025-05-06; first version 2024-12-24).

Weng et al., 2026. Zhaotian Weng; Antonis Antoniadis; Deepak Nathani; et al. 2026. Group-Evolving Agents: Open-Ended Self-Improvement via Experience Sharing. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v1, 2026-02-04; first version 2026-02-04).

Winston et al., 2026. Winston, Caleb; Wang, Ron Yifeng; Mirhoseini, Azalia; et al. 2026. Agent JIT Compilation for Latency-Optimizing Web Agent Planning and Scheduling. Proceedings of the 43rd International Conference on Machine Learning (ICML 2026). [formal citation](#). Version read: [HTML](#) (v2, 2026-05-29; first version 2026-05-20).

Wong et al., 2026. Wong, Mike; Hsieh, Kevin; Nath, Suman; et al. 2026. Skim: Speculative Execution for Fast and Efficient Web Agents. Version read: [HTML](#) (v2, 2026-05-19; first version 2026-05-15).

X. Zhang et al., 2026. Xuan Zhang; Wenxuan Zhang; See-Kiong Ng; et al. 2026. Self-Evolving World Models for LLM Agent Planning. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v2, 2026-09-01; first version 2026-06-29).

Xia et al., 2025. Chunqiu Steven Xia; Zhe Wang; Yan Yang; et al. 2025. Live-SWE-agent: Can Software Engineering Agents Self-Evolve on the Fly?. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v3, 2025-11-24; first version 2025-11-17).

Xu et al., 2025. Haoran Xu; Jiacong Hu; Ke Zhang; et al. 2025. SEDM: Scalable Self-Evolving Distributed Memory for Agents. SEA @ NeurIPS 2025 (NeurIPS 2025 workshop; OpenReview venue 'SEA @ NeurIPS 2025 Poster', venueid NeurIPS.cc/2025/Workshop/SEA). [formal citation](#). Version read: [HTML](#) (v3, 2025-09-26; first version 2025-09-11).

Xu et al., 2026. Wujiang Xu; Jiaojiao Han; Minghao Guo; et al. 2026. AEL: Evolving Agent Harness in Open-Ended Environments. EMNLP 2026, Findings track (Findings of ACL: EMNLP 2026), paper 5020-FIND, presentation type 'Not Presenting'; not the EMNLP main conference; Findings volume not yet in the ACL Anthology. [formal citation](#). Version read: [HTML](#) (v2, 2026-08-25; first version 2026-04-23).

Y. Li et al., 2026. Yanzhou Li; Yiran Zhang; Xiaoyu Zhang; et al. 2026. CODESKILL: Learning Self-Evolving Skills for Coding Agents. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v2, 2026-09-25; first version 2026-05-25).

Y. Yang et al., 2026. Yifan Yang; Ziyang Gong; Weiquan Huang; et al. 2026. SkillOpt: Executive Strategy for Self-Evolving Agent Skills. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v2, 2026-05-25; first version 2026-05-22).

Y. Yang et al., 2026. Yunhao Yang; Neel P. Bhatt; Kevin Wang; et al. 2026. VASO: Formally Verifiable Self-Evolving Skills for Physical AI Agents. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v1, 2026-06-03; first version 2026-06-03).

Y. Yang et al., 2026. Yang, Yanting; Jin, Can; Zhao, Jinman; et al. 2026. Act More, Decide Less: Skill-Guided Adaptive Action Chunking for Long-Horizon LLM Agents. Findings of the Association for Computational Linguistics: EMNLP 2026. Version read: [HTML](#) (v1, 2026-09-02; first version 2026-09-02).

Y. Zhang et al., 2026. Yuxuan Zhang; Cong Wei; Penghui Du; et al. 2026. RewardHarness: Learning Human Preferences for Image Editing with Only 100 Demonstrations. COLM 2026 (Third Conference on Language Modeling), main conference, poster (Poster Session 4, Imperial Ballroom #55); OpenReview venueid colmweb.org/COLM/2026/Conference, publication date 2026-07-08. [formal citation](#). Version read: [HTML](#) (v1, 2026-05-09; first version 2026-05-09).

Yadouni et al., 2026. Salma El Yadouni; Guanyi Li. 2026. TraceCompiler: Skill-Guided Mining and Compilation of LLM Agent Traces into Mostly Deterministic Workflows. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v1, 2026-08-03; first version 2026-08-03).

Yan et al., 2026. Zhiling Yan; Dingjie Song; Hanrong Zhang; et al. 2026. OpenSkill: Open-World Self-Evolution for LLM Agents. EMNLP 2026 (The 2026 Conference on Empirical Methods in Natural Language Processing), main conference, paper ID 3199-MAIN, virtual presentation (session 'Virtual 2', 2026-10-25, 18:00-19:30 CET, presenter Zhiling Yan), per the official detailed program linked from 2026.emnlp.org/program/. The ACL Anthology EMNLP 2026 volume is not yet published. [formal citation](#). Version read: [HTML](#) (v1, 2026-06-04; first version 2026-06-04).

Yin et al., 2025. Xunjian Yin; Xinyi Wang; Liangming Pan; et al. 2025. Gödel Agent: A Self-Referential Agent Framework for Recursively Self-Improvement. ACL 2025: Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), main conference, pages 27890-27913, Vienna, Austria, July 2025; Anthology ID 2025.acl-long.1354; DOI 10.18653/v1/2025.acl-long.1354. [formal citation](#). Version read: [HTML](#) (v4, 2025-05-31; first version 2024-10-06).

References (5/5)

Yuan et al., 2024. Lifan Yuan; Yangyi Chen; Xingyao Wang; et al. 2024. CRAFT: Customizing LLMs by Creating and Retrieving from Specialized Toolsets. International Conference on Learning Representations 2024 (ICLR 2024), main conference, poster (decision 'Accept (poster)'; OpenReview G0vdDSt9XM). [formal citation](#). Version read: [HTML](#) (v2, 2024-03-13; first version 2023-09-29).

Yuan et al., 2026. Boqin Yuan; Yue Su; Renchu Song; et al. 2026. ClawTrace: Cost-Aware Tracing for LLM Agent Skill Distillation. Agent Skills '26 Workshop at ACM CAIS 2026 (ACM Conference on AI and Agentic Systems), San Jose, May 26, 2026; OpenReview venue 'AgentSkills 2026 Oral'; workshop is non-archival. [formal citation](#). Version read: [HTML](#) (v2, 2026-05-25; first version 2026-04-26).

Yuksekgonul et al., 2025. Mert Yuksekgonul; Federico Bianchi; Joseph Boen; et al. 2025. Optimizing generative AI by backpropagating language model feedback. Nature, vol. 639, issue 8055, pp. 609–616 (2025), Article; published online 19 March 2025 (received 12 June 2024, accepted 16 January 2025). [formal citation](#). Version read: [HTML](#); [PDF](#) (v1, 2024-06-11; first version 2024-06-11).

Z. Liu et al., 2026. Zewen Liu; Zhan Shi; Yisi Sang; et al. 2026. Adaptive Auto-Harness: Sustained Self-Improvement for Agentic System Deployment on Open-Ended Task Streams. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v2, 2026-06-03; first version 2026-06-01).

Z. Wu et al., 2024. Zhiyong Wu; Chengcheng Han; Zichen Ding; et al. 2024. OS-Copilot: Towards Generalist Computer Agents with Self-Improvement. LLMAgents @ ICLR 2024 Poster: ICLR 2024 Workshop on Large Language Model Agents. [formal citation](#). Version read: [HTML](#) (v2, 2024-02-15; first version 2024-02-12).

Zelikman et al., 2024. Eric Zelikman; Eliana Lorch; Lester Mackey; et al. 2024. Self-Taught Optimizer (STOP): Recursively Self-Improving Code Generation. COLM 2024 (Conference on Language Modeling, first edition), main conference, Spotlight. [formal citation](#). Version read: [PDF](#) (v3, 2024-08-16; first version 2023-10-03).

Zhang et al., 2026a. Jenny Zhang; Shengran Hu; Cong Lu; et al. 2026. Darwin Gödel Machine: Open-Ended Evolution of Self-Improving Agents. ICLR 2026 (International Conference on Learning Representations 2026), main conference (proceedings track 'Conference'), poster. [formal citation](#). Version read: [HTML](#) (v3, 2026-03-12; first version 2025-05-29).

Zhang et al., 2026b. Jenny Zhang; Bingchen Zhao; Wannan Yang; et al. 2026. Hyperagents. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v1, 2026-03-19; first version 2026-03-19).

Zhang, 2026. Di Zhang. 2026. AgentDevel: Reframing Self-Evolving LLM Agents as Release Engineering. arXiv preprint. [formal citation](#). Version read: [HTML](#); [PDF](#) (v1, 2026-01-08; first version 2026-01-08).

Zhao et al., 2024. Andrew Zhao; Daniel Huang; Quentin Xu; et al. 2024. ExpeL: LLM Agents Are Experiential Learners. Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 38 No. 17: AAAI-24 Technical Tracks 17, AAAI Technical Track on Natural Language Processing II, pp. 19632-19642 (published 2024-03-24), DOI 10.1609/aaai.v38i17.29936. [formal citation](#). Version read: [HTML](#) (v3, 2024-12-20; first version 2023-08-20).

Zhao et al., 2025. Zhiyu Zhao; Haoxuan Li; Haifeng Zhang; et al. 2025. Curious Causality-Seeking Agents in Open-ended Worlds (arXiv listing title for 2506.23068: 'Curious Causality-Seeking Agents Learn Meta Causal World'). NeurIPS 2025 (poster), main conference: Advances in Neural Information Processing Systems 38, Main Conference Track; DOI 10.52202/085713-5145. [formal citation](#). Version read: [HTML](#) (v3, 2025-10-26; first version 2025-06-29).

Zhao et al., 2026. Yao Zhao; Aditya Shanmugham; Swastik Roy; et al. 2026. EchoPath: Execution-Level Replayable Memory for GUI Agents. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v1, 2026-09-15; first version 2026-09-15).

Zheng et al., 2025. Boyuan Zheng; Michael Y. Fatemi; Xiaolong Jin; et al. 2025. SkillWeaver: Web Agents can Self-Improve by Discovering and Honing Skills. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v1, 2025-04-09; first version 2025-04-09).

Zhong et al., 2026. Hongbin Zhong; Fazle Faisal; Luis França; et al. 2026. ActionEngine: From Reactive to Programmatic Web Agents via State Machine Memory. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v2, 2026-09-28; first version 2026-02-24).

Zhou et al., 2025. Huichi Zhou; Yihang Chen; Siyuan Guo; et al. 2025. Memento: Fine-tuning LLM Agents without Fine-tuning LLMs. arXiv preprint. [formal citation](#). Version read: [HTML](#) (v2, 2025-08-25; first version 2025-08-22).

Zhuge et al., 2024. Mingchen Zhuge; Wenyi Wang; Louis Kirsch; et al. 2024. GPTSwarm: Language Agents as Optimizable Graphs. ICML 2024: Proceedings of the 41st International Conference on Machine Learning, PMLR 235:62743-62767 (2024), main conference, Oral; ICML 2024 official paper data give decision 'Accept (Oral)', Oral 6A Agents and World Modeling, plus Poster Session 6. [formal citation](#). Version read: [HTML](#) (v3, 2024-08-22; first version 2024-02-26).

A1 · How the three columns were judged, and the totals per class

CLASS	WORKS	TIME OR MONEY OF RUNS (YES · PARTLY)	LEARNING COST (YES · PARTLY)	UNSEEN TASKS (YES · PARTLY)
Prompts and context	15	1 · 2	5 · 5	7 · 6
Experience memory	26	5 · 15	5 · 4	11 · 14
Skills and tools	34	9 · 14	6 · 11	8 · 23
Whole harness	30	8 · 4	7 · 16	5 · 23
Predicting what an action does	3	1 · 1	1 · 2	1 · 2
Descriptions, facts and preconditions	13	6 · 4	2 · 3	3 · 8
Page and place structure	7	5 · 2	1 · 3	1 · 6
Exploring and practising	9	3 · 2	1 · 3	4 · 4
Environments that check the final state (no learning)	3	1 · 2	0 · 0	3 · 0
Direction 1, each work once	93	21 · 28	19 · 33	28 · 57
Direction 2, each work once (final-state checkers excluded)	29	14 · 8	5 · 10	8 · 18

Time or money of runs — yes: measured seconds or dollars of executing tasks with what was learned; partly: tokens, steps, calls or an author’s estimate only. **Learning cost** — yes: a measured amount for learning, building or maintaining (tokens, dollars, time, compute or rollouts); partly: an estimate, iteration counts only, or a part knowingly left out. **Unseen tasks** — yes: tested on tasks that took no part in learning or selection; partly: an online stream that learns as it goes, the same workflow with new parameters, or an unstated split. These are our initial judgments from the cells of the per-work table (calc.), not re-read from the full texts; the representative works on pages 5 and 7 are to be checked one by one before the deck is final. Page 4 counts a different corpus: 178 agent methods in the calibrated set of 383 works.

A02 · Prompts and context (1/2)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
Reflexion (Shinn et al., 2023)	NeurIPS 2023	Text reflection added to next attempt on same task; model not updated	ALFWorld, QA, code; ALFWorld retried in same environment	ReAct GPT-3 prompt setting; ALFWorld cumulative 130 of 134 tasks; WebShop no improvement by four trials	Report 12 consecutive trials; tokens, cost, time not reported; trial count ≠ steps	○ ○ ○
Self-Refine (Madaan et al., 2023)	NeurIPS 2023	Current output and feedback history for next round	Text rewriting, code, math; per-question testing with answer changes	GPT-4 code optimization 27.3% → 36.0%; GPT-3.5 math 64.1 → 64.1; compared with same model single generation	Iteration limit 4; tokens, cost, time not reported	○ ○ ○
TextGrad (Yuksekgonul et al., 2025)	Nature 2025	Text variables; cross-question prompts or current-question solutions	Math, QA, code; prompt experiment split train/validation/test	gpt-3.5-turbo-0125 execution, gpt-4o feedback; Object Counting 77.8 → 91.9 vs DSPy 84.9; result table does not clearly mark split assignment	Prompt optimization batch size 3, 12 iterations; 1+1+1 model calls per code round; time and cost not measured	○ ● ●
MIPRO (Opsahl-Ong et al., 2024)	EMNLP 2024	Module instructions and examples	Multi-hop QA, evidence retrieval, classification; final held-out test	Llama-3-8B HotPotQA test exact match 46.4 vs unoptimized 36.1, mean of 5 runs; Iris 88.6 vs Bootstrap RS 94.1	Only trial budget evaluated; actual measured tokens, total calls, time, cost not reported	○ ● ●
ProTeGi (Pryzant et al., 2023)	EMNLP 2023	Prompt text	Classification task; train, dev, test separated, but final candidate selection by test F1 not clarified	Default gpt-3.5-turbo; Jailbreak test F1 when matching total API query budget: single selection after search 0.80, greedy 0.82, multi-candidate search 0.85; ~3 steps to peak	Author reported optimization sometimes past 1 hour, not per-setting timed table; deployment cost and time not measured	○ ● ●
Semantic Backpropagation (W. Wang et al., 2024)	arXiv	Fixed text instructions in computational graph	GSM8K, BBH, BigCodeBench; main comparison uses held-out test	gpt-4o-mini execution, gpt-4-turbo feedback update: GSM8K 93.2 vs TextGrad 78.2; BigCodeBench 27.8 vs 27.6; run count not reported	LIAR forward input tokens 273,300 vs removed neighbor info 289,850; optimization process metric, time and cost not measured	○ ● ●
Trace / OptoPrime (Cheng et al., 2024)	NeurIPS 2024	Prompts, function code or text-representable parameters	BBH held-out questions, controlled held-out initial conditions, new chess positions	BBH with reasoning template overall 78.6 vs DSPy prompt optimization 71.6; optimizer GPT-4-0125-Preview, in-program model not clearly stated	Word Sorting vs TextGrad: total experiment time 82.5 vs 211.1 min, including optimization, validation and testing; not per-question deployment latency	○ ● ●
PROMST (Chen et al., 2024)	EMNLP 2024	Action prompts; separately train Longformer scorer, not all models frozen	Web, text environments, planning; main performance independence on held-out not clarified	GPT-4 average progress score 0.69 vs PromptAgent 0.61; WebArena 0.62 vs APO 0.64; progress score ≠ full-task success rate	Time, cost, tokens or total calls not measured; some tasks with higher score need more steps, collisions also increase	○ ○ ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

A02 · Prompts and context (2/2)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
Dynamic Cheatsheet (Suzgun et al., 2026)	EACL 2026	Online-maintained text, strategies and code snippets	Math, science QA; accumulated in single test stream	GPT-4o Game of 24: baseline 10%, fixed-prompt zero memory 19%, retrieved organized memory 99%; Claude 3.5 Sonnet baseline 12% → 14%	AIME 2024 Claude: 370 / 494 / 1035 / 1831 tokens; memory version higher; token types and management overhead range unclear; time and cost not measured	
GPTSwarm (Zhuge et al., 2024)	ICML 2024 main conference	Node prompts, inter-agent connection probability; not LLM weights	GPT-3.5/GPT-4; MMLU, Crosswords, HumanEval	Crosswords same 20 questions optimization evaluation 0.465 → 0.575; HumanEval online 0.76 → 0.88	MMLU optimization 5.32 USD, inference 1.82 USD; HumanEval with optimization total 28.46 USD vs 1.61 USD, cannot be viewed as same-scope deployment cost	
AvaTaR (S. Wu et al., 2024)	NeurIPS 2024	Instructions and action sequences; task-direct action program reuse	Knowledge base retrieval, multi-tool QA; training then testing official and human held-out queries	claude-3-opus STaRK Amazon/MAG/Prime top-1 hit rates 49.87 / 44.36 / 18.44 vs Reflexion 42.79 / 40.71 / 14.28	Time and cost not measured; timeout-triggered rewrite is mechanism only, threshold not given	
DSPy (Khattab et al., 2024)	ICLR 2024	Select prompts from successful trajectories and ensemble; fine-tuning branch not counted	GSM8K, HotPotQA; held-out test present, some report dev set only	GPT-3.5 GSM8K uncompiled vanilla test 25.2 vs CoT bootstrap+ensemble 81.6; architecture and prompt change together, not single factor	Compilation needs minutes to tens of minutes per author estimate; deployment tokens, time, cost not measured	
GEPA (Agrawal et al., 2026)	ICLR 2026	Module prompts only, model frozen	QA, instructions, privacy rewriting, math; train validation test separated	Qwen3 8B AIME-2025: 32.00 vs MIPROv2 20.00, GRPO 38.00; overall advantage ≠ every task wins	Per-task optimization rollout total 6871 / 3593 / 7051 / 2426 / 1839 / 1839; GRPO per item 24000; not model calls, time or cost	
ACE (Q. Zhang et al., 2026)	ICLR 2026	Text context entries, model frozen	AppWorld, finance; offline held-out and online task stream tested separately	DeepSeek-V3.1 AppWorld: offline labeled 59.4, unlabeled 57.2, baseline 42.4; no warm-up online 56.1; FiNER unlabeled online 67.3 < baseline 70.7	AppWorld offline adaptation 9517 s vs GEPA 53898 s; online FiNER 2.9 vs DC 17.7 dollars; adaptation cost metric; statistical scope has unexplained differences	
ESPO (L. Liu et al., 2026)	EMNLP 2026 Findings (listed in the programme)	Natural-language module prompt	Seven text benchmark, each 500 held-out; Claude Sonnet 4.5	Main optimize single average 74.67% vs GEPA 70.91%; prompt 1,004 vs 1,878 character	PUPA optimize ~3.4M tokens, 2h 13m; no deployment acceleration measured	

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ◐ partly, ○ no (initial judgment, calc.; Appendix A1).

A03 · Experience memory (1/4)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
Dynamic Cheatsheet (Suzgun et al., 2026)	EACL 2026	Online-maintained text, strategies and code snippets	Math, science QA; accumulated in single test stream	GPT-4o Game of 24: baseline 10%, fixed-prompt zero memory 19%, retrieved organized memory 99%; Claude 3.5 Sonnet baseline 12% → 14%	AIME 2024 Claude: 370 / 494 / 1035 / 1831 tokens; memory version higher; token types and management overhead range unclear; time and cost not measured	
ACE (Q. Zhang et al., 2026)	ICLR 2026	Text context entries, model frozen	AppWorld, finance; offline held-out and online task stream tested separately	DeepSeek-V3.1 AppWorld: offline labeled 59.4, unlabeled 57.2, baseline 42.4; no warm-up online 56.1; FiNER unlabeled online 67.3 < baseline 70.7	AppWorld offline adaptation 9517 s vs GEPA 53898 s; online FiNER 2.9 vs DC 17.7 dollars; adaptation cost metric; statistical scope has unexplained differences	
Agent Workflow Memory (AWM) (Wang et al., 2025a)	ICML 2025	Infer text workflows from trajectories; action encapsulation has variants	WebArena online; Mind2Web cross-task, site, domain	GPT-4-0613 WebArena: 35.5 vs BrowserGym 23.5 success rate; compared input contains HTML, not strict same interface; Mind2Web full-task success rate 0.7—4.8	WebArena average steps 5.9 vs author rerun with accessibility-tree-only BrowserGym 7.9; step comparison not same as 23.5 success-rate row	
ExpeL (Zhao et al., 2024)	AAAI 2024	Suggestion list and retrieved successful trajectories	HotpotQA, ALFWorld, WebShop; train and eval tasks separated	gpt-3.5-turbo-0613 ALFWorld single-attempt success 59% vs Reflexion after third retry 54%; different test budget	Training retry limit 3 only; extraction calls, tokens, time or cost not measured	
MemRL (S. Zhang et al., 2026)	arXiv	Experience repository and retrieval value based on reward	Code, OS, database, ALFWorld, HLE; repeated training vs frozen memory with held-out separated	Held-out average success rate vs MemP +2.8 pp; ALFWorld +5.8, OS +2.6; different baseline foundation models	HLE Gemini-3-pro average ~32K tokens per question across 10 epochs; whether measured per attempt unclear; no comparable MemP figures	
MemQ (Liao et al., 2026)	arXiv	External experience value and generated relevance graph	Multimodal QA, code, tools, OS; freeze best round memory for held-out	Gemma-4-E4B-it LiveCodeBench: 51.43±2.86 vs RAG 50.48±1.65, MemRL 45.71±2.86; 3 seeds; best round selection not clarified	Time, cost, tokens or calls not measured; 'faster learning' means per-round success rate	
AEL (Xu et al., 2026)		Main config learns memory retrieval selection, reflection, rules; code evolution default off	Claude Haiku 4.5 multi-task allocation, Gemini 3.1 Flash-Lite work-order stream	Multi-task allocation test Sharpe 2.13 +/- 0.47; EvoTool 1.37 +/- 1.74, but difference p=0.39; planning evolution -1.72	Author estimate full run \$2.80, prediction ~\$2, reflection evolution ~\$0.80; latency not measured	
Metis (Dai et al., 2026)	arXiv	Environment facts, pitfalls, plans and callable code from repeated plans	AppWorld; offline train once, freeze memory then held-out test	GPT-4o official test TGC 60.1 vs no memory 51.8; resampled test 66.1 vs 54.8	Official per-question execution 97.4K vs 112.6K tokens, 11.25 vs 14.55 turns; does not include Memory Manager; build 7.9M tokens without trajectory collection	

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ◐ partly, ○ no (initial judgment, calc.; Appendix A1).

A03 · Experience memory (2/4)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
Mem²Evolve (Cheng et al., 2026)	ACL 2026	Expert agents, executable tools and experience	General assistant, math, QA, planning, embodied, shopping; main result test stream accumulation inferred as entries	GPT-5-chat ALFWorld 94.31 vs AFLOW 93.40, HotpotQA 60.80 same as AFLOW; cross-task memory analysis has separate held-out	Tool average repair rounds 1.01 → 0.48; tokens, time cost not measured; rounds not convertible to them	● ○ ●
FORGE (Bogdanov et al., 2026)	ACM CAIS 2026	Multiple individual rules or example memory, champion broadcasts entire	CybORG CAGE-2 fixed attacker; train then freeze evaluation, same scenario	Gemini best example scheme reward -24.5 vs zero-shot -189.6, closer to zero is better; other attackers not tested	Gemini rule scheme per-individual adaptation 7.3M +/- 1.6M tokens vs Reflexion 11.4M +/- 0.5M; time and cost not measured	○ ● ●
G-Memory (G. Zhang et al., 2025)	NeurIPS 2025	Multi-agent interaction, task and suggestion associative memory	QA, text embodied, planning; independent memory build vs held-out not clear	Qwen-2.5-14b + MacNet ALFWorld 58.21% → 79.10%; run count not reported	AutoGen+PDDL relative to no memory 1.4×10^6 tokens more, score 10.32 higher; management cost not separated; time and cost not measured	● ○ ●
CTIM-Rover (Lindenbauer et al., 2025)	ACL 2025 REALM workshop	General and repo-level text suggestions, can include successful trajectory examples	SWE-bench Verified, 401 train, 45 test, also Lite trajectories	GPT-4o o1 extraction: full 40% vs AutoCodeRover 42%; text memory only 31%; whether extra Lite trajectories overlap not clarified	Average turns 11.16 → 9.62; final context 11414.02 → 17544.93 tokens, not cumulative tokens; time and cost not measured	● ○ ●
EvolveMem (J. Liu et al., 2026)	arXiv	Retrieval config, answer style and re-extracted memory for gaps	LoCoMo conversational QA, MemBench; main performance held-out independence not reported	GPT-4o LoCoMo F1 0.543 vs SimpleMem 0.432; evolution same question 30.5 → 54.3	Single LoCoMo sample full evolution 25-35 min; retrieval 15 ms, validation ~2-3 s additional; no end-to-end latency	● ● ●
DecentMem (Hao et al., 2026)	arXiv	Per-agent local experience pool and exploration/reuse router	Math, code, BBH, ALFWorld; held-out and retry count not reported	Qwen3-4B + AgentNet average accuracy 53.69 vs G-Memory 43.37, no memory 35.20	Qwen3-8B BBH vs G-Memory token reduction 32% / 49% / 47%, unclear if judging included; no-memory still cheaper	● ○ ●
EXG (Jin et al., 2026)	arXiv	Failure-success experience graph, connecting failures to repair cases	Code and multi-hop QA; online stream and frozen graph 7:3 held-out	Qwen3-Coder-Flash HumanEval held-out: EXG-Reflexion 0.879 vs ExpeL 0.909; online has larger gains	HumanEval baseline EXG: 1.20 vs Reflexion 1.83 model calls; LLM latency 3,259 vs 3,416 ms; total tokens 125,304 vs 117,378; cost-graph model not clear	● ○ ●
Meta-TTL (Lou et al., 2026)		Cross-round action prompts for same task and 'how to change prompt' prompt learned offline	Jericho, WebArena-Lite, tau2-bench; new instances and new domains	Gemini-3-Flash action, GPT-5 adaptation; WebArena-Lite unseen domain W-AUC 0.766 vs TextGrad 0.728, 3 runs, not single success rate	Jericho per session 860.9 s vs Static 593.2 s; 204.0 vs 231.0 calls; 364k vs 196k tokens; offline ~\$30-70/benchmark	● ○ ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

A03 · Experience memory (3/4)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
SEDM (Xu et al., 2025)	NeurIPS 2025 SEA workshop	Injectable memory fragments with utility weight and source	LoCoMo, FEVER, HotpotQA; independent build vs evaluation not clear	GPT-4o-mini FEVER: 66 vs G-Memory 62, no memory 57; HotpotQA 39 / 38 / 34	FEVER prompt tokens 2.47M vs G-Memory 3.62M, no memory 1.65M; replay cost and time cost not measured	● ○ ●
Agent S (Agashe et al., 2025)	ICLR 2025	Frozen model, update full-task and sub-task text memory; interface layer fixed	GPT-4o/Claude-3.5 OSWorld/WindowsAgentArena; ongoing evaluation update after exploration	GPT-4o OSWorld 20.58% vs literature baseline 11.21%; ablation subset full 26.15% vs no pre-exploration 15.38%, no independent frozen-memory held-out	Checked entry does not report steps, time, tokens, cost or build cost	○ ○ ●
SE-Agent (Guo et al., 2025)	NeurIPS 2025	Candidate trajectory pool for same task, rewriting, recombining and filtering	SWE-bench Verified; per-question search, not cross-question held-out memory	GPT-4o: Pass@1 40.4% vs SWE-Agent 22.4%, SWE-Search 32.6%; internal multi-candidate, not single trajectory	Default 10 candidates; max API cost graph unit unclear; actual tokens, time cost not measured	○ ○ ○
WebCoach (Liu et al., 2025)	ICLR 2026 MemAgents workshop	External trajectory summary, additional Coach injects suggestions to action model	WebVoyager 643 real-time web tasks; dynamic library accumulates same-stream same-task but excludes same-task-ID retrieval	Skywork-38B + Qwen3-8B Coach: 0.614 vs 0.473; Qwen-VL-7B main table 0.311 vs 0.328	Skywork average per question 215 → 395 s, actions 10.7 → 10.2; time includes extra inference; no token cost measurement	● ○ ●
ReasoningBank (Ouyang et al., 2026)	ICLR 2026	Text experience pool extracted from success and failure trajectories	WebArena, SWE-bench; online single-task stream	WebArena Gemini-2.5-flash success rate 40.5 → 48.8; Shopping judge accuracy 72.7%	WebArena average ReAct steps 9.7 → 8.3; another table total tokens 50,847.4 → 53,054.5, baseline foundation not specified; time and cost not measured	● ○ ●
ReAP (Azam et al., 2025)	arXiv	Text reflection retrieved by task similarity	WebArena, same-task redo vs similar new-task tested separately	AgentOccam gpt-4o-2024-08-06; 80% build, 20% test new-task setup; reflection vs no memory +11 points; old success task best only 84%	Same-task redo average steps 11.92 → 10.08; new task step reduction 29.1%; time cost not measured; build reports one trajectory per task only	● ● ●
Memento (Zhou et al., 2025)	arXiv	Case library; parameter branch additionally trains two-layer MLP retrieval scorer	GAIA and research QA; GAIA validation set builds library then test set	GPT-4.1 planning, o3 execution: GAIA validation Pass@3 87.88%, test set using only validation-accumulated library 79.40%; different protocols	Each level sample 10 questions; GAIA L1 input/output 26k/4.7k, L3 121k/9.8k tokens; not complete learning cost, no savings comparison	● ○ ●
MobileGPT (Lee et al., 2024)	ACM MobiCom 2024	Per-app page-subtask graph and action sequence	Mobile apps; cold start then warm start same task with parameter change	GPT-4-turbo; with human repair ablation warm start 98.75%; main table without human repair cannot mix	Human-repair path warm start vs same prompt Derive-only: latency -62.5%, cost -68.8%; offline explore per app 10 to 15 minutes, total \$10.78	● ● ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ○ partly, ○ no (initial judgment, calc.; Appendix A1).

A03 · Experience memory (4/4)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
Grounding Agent Memory (Suresh et al., 2026)	arXiv	Only modify post-task experience library, add environment read-only probe	CLBench 40 question, unreported structure change; gpt-5.4; five paired run	Pass 39% → 73%; executor per-question SQL query 8.8 → 4.7	Full execution \$3.38 → \$1.68; no experience-build and probe-maintain	● ○ ●
Are Online Skill and Memory Modules Always Worth Their Tokens? (Hajimiri et al., 2026)	EMNLP 2026 main conference (listed in the programme)	Control AWM, ASI, ReasoningBank vs plain executor budget config	WebArena four domain, Gemini 3 Flash, three run	Vanilla-IB success 44.78%, 73.6K total tokens/question, that model four-config best simultaneously	Token only; plain 15 step+pruning, enhance 10 step, near-equal budget	● ● ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

A04 · Skills and tools (1/4)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
Metis (Dai et al., 2026)	arXiv	Environment facts, pitfalls, plans and callable code from repeated plans	AppWorld; offline train once, freeze memory then held-out test	GPT-4o official test TGC 60.1 vs no memory 51.8; resampled test 66.1 vs 54.8	Official per-question execution 97.4K vs 112.6K tokens, 11.25 vs 14.55 turns; does not include Memory Manager; build 7.9M tokens without trajectory collection	● ○ ●
Mem²Evolve (Cheng et al., 2026)	ACL 2026	Expert agents, executable tools and experience	General assistant, math, QA, planning, embodied, shopping; main result test stream accumulation inferred as entries	GPT-5-chat ALFWorld 94.31 vs AFLOW 93.40, HotpotQA 60.80 same as AFLOW; cross-task memory analysis has separate held-out	Tool average repair rounds 1.01 → 0.48; tokens, time cost not measured; rounds not convertible to them	● ○ ●
Agent S (Agashe et al., 2025)	ICLR 2025	Frozen model, update full-task and sub-task text memory; interface layer fixed	GPT-4o/Claude-3.5 OSWorld/WindowsAgentArena; ongoing evaluation update after exploration	GPT-4o OSWorld 20.58% vs literature baseline 11.21%; ablation subset full 26.15% vs no pre-exploration 15.38%, no independent frozen-memory held-out	Checked entry does not report steps, time, tokens, cost or build cost	○ ○ ●
SkillWeaver (Zheng et al., 2025)	arXiv	Playwright function library from web exploration, includes usage notes and page prerequisites	GPT-4o WebArena; self-practice then test benchmark tasks	Success rate 22.6 → 29.8; online human-judged subset 40.2 → 56.2, Cars 11.1 → 11.1	Per site 160 exploration iterations; checked entry does not report runtime and exploration time cost	○ ● ●
Voyager (G. Wang et al., 2024)	TMLR 2024	Executable Minecraft skills, completed and failed task tables	GPT-4; MineDojo/Mineflayer; exploration vs new worlds no task separation	New world each unseen task 3/3 success; ReAct, Reflexion, AutoGPT 0/3 within 50 prompting iterations	'15.3× faster' is prompt-iteration ratio to unlock wood tools; seconds, tokens, dollars checked entry not reported	● ○ ●
MetaAgent (Qian et al., 2025)	arXiv	Frozen model, accumulate text experience and raw tool result knowledge base; no explicit environment-rules model	QwQ-32B GAIA/WebWalkerQA/BrowseComp; pre-fill library with same-batch task simulation	WebWalkerQA 52.1 vs WebThinker-32B-RL 46.5; GAIA 47.6 below its 48.5; no held-out split	Checked entry does not report measured cost or time	○ ○ ○
SkillOpt (Y. Yang et al., 2026)	arXiv	Deploy skill documentation; execution model and framework fixed	GPT/Qwen; QA, table, document, math, ALFWorld; independent test	GPT-5.5 direct dialogue average 58.8 → 82.3 vs no skill; best per-cell baseline average 76.9	Training tokens e.g. SpreadsheetBench 21.4M; cost per question inconsistent scope; deployment per-task usage and time not reported	○ ● ●
VASO (Y. Yang et al., 2026)	arXiv	Logic skill rules, proposition mapping, plan templates	Jackal/PX4 planning and simulation; GPT-5-nano skill generation, GPT-4o-mini planning	Automated rules satisfy 96.8, task completion 86.5; LLM-Planner 90.3/89.3; new prompt safety 92–100%, completion 80–96%	Author reported under 20 minutes per skill; skill execution time and cost not reported	○ ● ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

A04 · Skills and tools (2/4)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
AgentOrchestra (W. Zhang et al., 2025)	arXiv	Prompts, tool/agent/memory/environment code and scheme version	GAIA, HLE, GPQA, AIME; mainly Gemini 3 Flash Preview, browse component different models	HLE 55.2 → 59.6; GAIA Validation 89.70 → 93.33; independent evolution split not stated	Author rough estimate simple task 30 seconds/5k tokens, complex 10 minutes/100k; no Vanilla/Evolved cost comparison	● ○ ●
GenericAgent (Liang et al., 2026)	arXiv	SOP, script, hierarchical memory; fixed tool interface	Multiple models; local task, web, repeated download and research tasks	Same GPT-5.4 repeated download 102 seconds → ~66 seconds; not held-out; five long tasks success 100%, vs Claude Code 100%, OpenClaw 80%	Repeated download tokens 200,439 → 100,000; five long tasks 220.8 s vs OpenClaw 183.1 s; evolution cost not separated	● ● ●
STELLA (Jin et al., 2025)	arXiv	Biomedical tool pool, reasoning template	Claude 4 Sonnet + Gemini 2.5 Pro; HLE biomedical, LAB-Bench subset	HLE 14% → 26%; LitQA ~52% → 63% with trial budget increase; clear evolution/held-out separation missing	Only trials 1x to 9x reported; time/token/cost not reported	○ ● ●
DRAFT / From Exploration to Mastery (Qu et al., 2025)	ICLR 2025	Rewrite natural-language tool documentation, do not train execution model	GPT-4o writes docs, GPT-4o/mini/Llama-3-70B use; RestBench, ToolBench	GPT-4o TMDb correct-path rate 88.00 vs ReAct 71.00; tool path includes standard subsequence, not final-state check	Checked entry does not report run and improvement call total, tokens, time, cost	○ ○ ●
CODESKILL (Y. Li et al., 2026)	arXiv	Text skill library + independently-trained SFT/RL management model	Frozen Qwen3.5-35B-A3B or GPT-5.4-mini; programming and environment repair	Qwen execution policy average 29.32 → 40.35; skill library built from test-set prerun, excluding current-task source, not strict online stream	Average reasoning steps 44.12 → 35.99; success-task scope inconsistent; manager RL ~210 hours, SFT ~20 hours	● ● ●
LATM (Cai et al., 2024)	ICLR 2024 main conference	Python tools + call examples + tool cache	GPT-4 create; GPT-3.5 Turbo/GPT-4 use; logic, bracket, sorting, schedule held-out	GPT-3.5 Turbo Dyck Language 20.4 → 92.2; GPT-4 Logical Deduction 88.8 → 86.6	Only O(nc+C) parse cost and call-budget design; time/tokens/dollars not measured	○ ● ●
Alita-G (Qiu et al., 2025b)	arXiv	Successful-trajectory tools abstracted into MCP toolbox, retrieval select	Claude-Sonnet-4 manage + GPT-4.1 browse; GAIA validation set build, retest	GAIA pass@1 75.15% → 83.03%; PathVQA/HLE build-test relationship not explained	GAIA average generated tokens 12,305 → 10,394; build cost and latency not measured	● ○ ○
OS-Copilot / FRIDAY (Z. Wu et al., 2024)	ICLR 2024 LLMagents workshop	Frozen model, update program tool library; semantic memory is design component, experiment mainly validates tools	GAIA dev set learn, private test; self-created table tasks then SheetCopilot-20 test	GPT-4 SheetCopilot-20 Pass@1 60% vs no learning 0%; GAIA Level 3 both 6.12%	GAIA Level 1 table average time 118 vs 124 s; Level 2 221 vs 239 s; no-learning cost	● ○ ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

A04 · Skills and tools (3/4)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
OpenAgent (Lyu et al., 2025)	ACL 2025 main conference	GitHub repo integration, Docker environment, experience summary	GPT-4, GPT-3.5; OpenAct; rerun experience experiment separate	GPT-4 51-question environment/experience reuse 47.0 → 82.3; main table first-time task 47.3 vs ReAct 24.6, average scope not fully recomputable	Time/tokens/dollars checked entry not reported	○ ○ ●
OpenSkill (Yan et al., 2026)		Per-target-task skill script + substitute test	Opus 4.6/Claude Code, GPT 5.2/Codex; SkillsBench; same-task skill build then rescore	SkillsBench no skill → OpenSkill 25.5% → 43.6%, 25.0% → 42.1%; 5 evals per skill	Opus average run 465.0 → 845.4 s; skill build ~749K tokens/39 min; per-stage includes estimates, totals not fully compatible	● ● ●
CoEvoSkills (H. Zhang et al., 2026)	COLM 2026 main conference	Skill documentation, code and co-improved substitute test	Opus 4.6/Claude Code, GPT-5.2/Codex; SkillsBench same-question improvement and test	Opus pass rate 30.6% → 71.1%, manual skill 53.5%; not held-out generalization	Average 4.1 validation cycle, 2.4 ground-truth round; seconds/tokens/dollars not measured	○ ● ●
AgentDistill (Qiu et al., 2025a)	arXiv	Extract and merge MCP toolbox from frozen teacher trajectories for frozen student	GPT-3.5, Qwen3-8B, LLaMA3.1-8B; Game of 24, medical VQA; held-out relationship not stated	GPT-3.5 Game of 24 accuracy 34.3±3.2 → 82.7±0.6; build 100 validation examples	Only tool-call proportion reported; call count, time, tokens, dollars not measured	● ○ ●
CRAFT (Yuan et al., 2024)	ICLR 2024 main conference	Abstract and retrieve Python functions from solutions	GPT-4 build tools, GPT-3.5 Turbo main execution; VQA, TabMWP, MATH algebra; train source and eval separate	GPT-3.5 TabMWP 80.6 → 88.4, but same toolbox BM25 is 89.2; GQA SAcc 35.0 → 45.4	Tool build author estimate ~\$2,500; runtime cost not measured	○ ● ●
ActionEngine (Zhong et al., 2026)	arXiv	State-action graph, locator and descriptor, online general-purpose repair write-back	WebArena 655 tasks, Claude Opus 4.6; template warm-up then eval, patcher off	91.2%; no warm-up graph 73.1%	27 s vs 87 s; \$0.05 vs \$0.40, both per-task relative Claude Code; no explore/warm-up; break-even 39-101 tasks and missing warm-up	● ● ●
WALT (Prabhu et al., 2026)	ICLR 2026 main conference	New action script, parameter constraint and tool description	WebArena, VisualWebArena; GPT-5 plan, GPT-5-mini execute	Average success 50.1% / 52.9%; same structure no-tool control best +30.7% relative, includes parser and verifier co-change	Steps best improve 1.4x; offline tokens, dollars, time not reported	● ● ●
SpeedRunner (Huang et al., 2026)	arXiv	Frozen-model executable skill library, add/remove/modify functions	ScienceWorld, BabyAI, Crafter; gpt-5.4-mini; 200 online task, 30 held-out, 3 seed	BabyAI from ReAct ~67% to near-perfect; Crafter all-method average achievement unlock <30%	BabyAI usage ~one eighth; output-tokens vs dollars graph scope conflict, learning total cost not reported	● ● ●
ASI (Wang et al., 2025b)	COLM 2025 main conference	Python skill add to action library	WebArena online stream, no separate train set; claude-3.5-sonnet	40.4% vs plain Claude 32.7%; average action 5.0 vs 5.6, one skill = one action	Learn tokens, dollars, time not reported; step scope not equal to raw operations	● ○ ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

A04 · Skills and tools (4/4)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
AppAgentX (Jiang et al., 2025)	arXiv	Page/element relation memory and high-level action	AppAgent 50 task/9 app; GPT-4o; five per-question	With chain memory then add action: success 70.8% → 71.4%; steps 9.1 → 5.7	Per-task tokens 9.26k → 4.94k; per-step time 23 → 16 s, successful-task-only, build cost unclear	● ○ ●
TraceCompiler (Yadouni et al., 2026)	arXiv	Data-dependency executable workflow	AppWorld one Venmo intent, three instance; author hand-translates execution	Same instance tool calls 34 → 11; hold-one-instance 15 of 21 state test pass	Tool call only; compile cost and net save not tested	● ○ ●
EchoPath (Zhao et al., 2026)	arXiv	Parameterized GUI trajectory memory, precondition and lifecycle	OSWorld-Verified 159 task with established memory; same-task second try change resolution, codex-gpt-5.5-medium	91.2% vs Synapse 91.8%	Median tokens 20,370 vs 586,386; time 127.5 s vs 315.7 s; first build median ~572k tokens/4.5 min counted success-only	● ● ○
SKILL.nb (Hattami et al., 2026)	ICML 2026 FAGEN workshop	Text step, code, pre/post check and version library, frozen model	WebArena-Verified 812 task, gpt-5.3-codex, shared framework different reasoning intensity	Single round 53.7%, +3.9 pp next; three perturbation reruns consistency 91.7%	Internal maintain/update 36k tokens per success, execution included unclear; no dollars or time	● ● ●
Are Online Skill and Memory Modules Always Worth Their Tokens? (Hajimiri et al., 2026)	EMNLP 2026 main conference (listed in the programme)	Control AWM, ASI, ReasoningBank vs plain executor budget config	WebArena four domain, Gemini 3 Flash, three run	Vanilla-IB success 44.78%, 73.6K total tokens/question, that model four-config best simultaneously	Token only; plain 15 step+pruning, enhance 10 step, near-equal budget	● ● ●
ClawTrace / CostCraft (Yuan et al., 2026)	ACM CAIS 2026 Agent Skills workshop, non-archival	Trajectory-summary exported skill document, frozen model	SpreadsheetBench 30 held-out, OpenClaw + gpt-5.4; also migrate SkillsBench	Remove delete patch after two-type sum degrade 5 → 14; migrate post cost median not decrease	Execution cost with cache; SkillsBench median \$0.143 vs \$0.144; extract cost not reported	● ○ ●
AXIS (Lu et al., 2025)	ACL 2025 main conference	Translate UI trajectory into API-code skill	Microsoft Word, 50-task feasibility; location entry incomplete learn/test split list	Relative UFO: 84.0% vs 52.0%, 2.0 vs 3.2 step	29.9 s vs 59.5 s; backend \$0.2 vs \$0.4; full learn cost location entry not reported	● ○ ●
Skim (Wong et al., 2026)	eligibility record: EuroSys 2027	URL template, answer structure and low-cost execution path	WebVoyager/WebShop, WebVoyager/AgentOccam/BrowserUse framework	Cost median 1.9x lower, latency -33.4%, reported accuracy no decrease	Per-task dollars and end-to-end time, include fail and retry; offline amortize location entry not reported	● ○ ●
Space (Y. Yang et al., 2026)	EMNLP 2026 Findings	Supervise policy model output multi-action block using skill boundary, deploy no skill library	ALFWorld, ScienceWorld; location entry does not list per-setting model and run count	Success improve 7.0%-31.3%, decision round best -78.9%; percentage meaning not itemized	Only decision round and training step given; no time or dollars saved	● ● ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ○ partly, ○ no (initial judgment, calc.; Appendix A1).

A05 · Whole harness (1/4)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
Trace / OptoPrime (Cheng et al., 2024)	NeurIPS 2024	Prompts, function code or text-representable parameters	BBH held-out questions, controlled held-out initial conditions, new chess positions	BBH with reasoning template overall 78.6 vs DSPy prompt optimization 71.6; optimizer GPT-4-0125-Preview, in-program model not clearly stated	Word Sorting vs TextGrad: total experiment time 82.5 vs 211.1 min, including optimization, validation and testing; not per-question deployment latency	○ ● ●
GPTSwarm (Zhuge et al., 2024)	ICML 2024 main conference	Node prompts, inter-agent connection probability; not LLM weights	GPT-3.5/GPT-4; MMLU, Crosswords, HumanEval	Crosswords same 20 questions optimization evaluation 0.465 → 0.575; HumanEval online 0.76 → 0.88	MMLU optimization 5.32 USD, inference 1.82 USD; HumanEval with optimization total 28.46 USD vs 1.61 USD, cannot be viewed as same-scope deployment cost	● ● ●
AEL (Xu et al., 2026)		Main config learns memory retrieval selection, reflection, rules; code evolution default off	Claude Haiku 4.5 multi-task allocation, Gemini 3.1 Flash-Lite work-order stream	Multi-task allocation test Sharpe 2.13 +/- 0.47; EvoTool 1.37 +/- 1.74, but difference p=0.39; planning evolution -1.72	Author estimate full run \$2.80, prediction ~\$2, reflection evolution ~\$0.80; latency not measured	○ ● ●
EvolveMem (J. Liu et al., 2026)	arXiv	Retrieval config, answer style and re-extracted memory for gaps	LoCoMo conversational QA, MemBench; main performance held-out independence not reported	GPT-4o LoCoMo F1 0.543 vs SimpleMem 0.432; evolution same question 30.5 → 54.3	Single LoCoMo sample full evolution 25-35 min; retrieval 15 ms, validation ~2-3 s additional; no end-to-end latency	● ● ●
SEDM (Xu et al., 2025)	NeurIPS 2025 SEA workshop	Injectable memory fragments with utility weight and source	LoCoMo, FEVER, HotpotQA; independent build vs evaluation not clear	GPT-4o-mini FEVER: 66 vs G-Memory 62, no memory 57; HotpotQA 39 / 38 / 34	FEVER prompt tokens 2.47M vs G-Memory 3.62M, no memory 1.65M; replay cost and time cost not measured	● ○ ●
GenericAgent (Liang et al., 2026)	arXiv	SOP, script, hierarchical memory; fixed tool interface	Multiple models; local task, web, repeated download and research tasks	Same GPT-5.4 repeated download 102 seconds → ~66 seconds; not held-out; five long tasks success 100%, vs Claude Code 100%, OpenClaw 80%	Repeated download tokens 200,439 → 100,000; five long tasks 220.8 s vs OpenClaw 183.1 s; evolution cost not separated	● ● ●
Symbolic Learning (Ou et al., 2025)		Prompts, tools, node flow; model weights fixed	GPT-3.5/GPT-4; QA, math, code, composition; split not explained	GPT-4 MATH 60.7 vs Agents 56.0; standard benchmark disables tools, cannot prove tool creation effective	Time and cost checked entry not reported	○ ○ ●
ADAS (Hu et al., 2025)	ICLR 2025 main conference	Fixed small framework forward() program	GPT-4o search, GPT-3.5 execution; ARC, QA reasoning held-out test	DROP F1 79.4 vs 65.8; MGSM 53.4 vs 39.0, each strongest hand-written baseline	One search + eval ~500 USD (ARC), 300 USD (reasoning); deployment per-question not measured	○ ● ●
MOSS (Cai et al., 2026b)	arXiv	OpenClaw code and tool-result mediation	DeepSeek V3.2; claweval same four-task failures and retest	Average grader score 0.2526 → 0.6100, 3 per-question; no held-out task	Only iteration-1 convergence reported; time/tokens/dollars not measured	○ ● ○

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

A05 · Whole harness (2/4)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
RewardHarness (Y. Zhang et al., 2026)	COLM 2026 main conference	Reward skill and tool-specification text library, vision model frozen	Qwen2.5-VL-7B/Gemini-2.0-Flash; image editing ranking	Qwen average 45.7 vs bare model 30.3, trained EditReward 42.0; do not treat downstream image-model training as no-training evidence	77 iteration scale only reported; tokens/seconds/dollars not measured	○●●
STOP (Zelikman et al., 2024)	COLM 2024 main conference	Python improver self-improve by calling fixed model	GPT-4, GPT-3.5, Mixtral; small algorithm problems; held-out and new-task transfer	GPT-4-selected improver transfer 3SAT 21.2% → 75.1%; GPT-3.5, Mixtral average degrade	LPN ~3 thousand GPT-4 calls per round each run; deployment time dollars not measured	○●●
Huxley-Gödel Machine / HGM (Wang et al., 2026)	ICLR 2026 main conference	Agent code + search-tree expand/eval schedule	SWE-bench, Polyglot; SWE GPT-5 modify, GPT-5-mini test	Same 800-eval budget SWE-Verified-60 40% → 56.7%; filter overlap SWE Lite held-out 34.8% → 40.1%	Optimization allocated CPU-hours: SWE 517 vs DGM 1231; not per-question deployment latency	○●●
Gödel Agent (Yin et al., 2025)	ACL 2025 main conference	Policy code and self-modifying program	GPT-4o optimize, GPT-3.5 bounded execution; QA/math held-out test	Bounded MGSM 64.2±3.4 vs Meta Agent Search 53.4±3.5; free version calls stronger model incomparable	Full evolution author estimate ~\$15; per-question execution cost and time not measured	○●●
JudgeFlow (Ma et al., 2026)	arXiv	Workflow logic blocks and configuration	gpt-4o-mini; math/code; train-test split present, candidate-scoring split not clear	MATH 58.5 vs MermaidFlow 55.4; MATH → GSM8K 92.89 vs AFlow 91.95	Single GSM8K round evaluation \$0.45, Judge \$0.01; deployment time and cost not measured	○●●
Adaptive Auto-Harness (Z. Liu et al., 2026)	arXiv	Task-type branching prompts, tools, infrastructure	Sonnet4.6 execute, Opus4.6 modify; prediction market, security challenge, event forecast online stream	CTF-Dojo 37.2 → 50.2; FutureX 31.0 → 47.3, but A-Evolve 47.5	FutureX average time 244.4 → 47.6 s; PolyBench 18.2 → 42.2 s; improver tokens not saved, evolution overhead not counted	●○●
Continual Harness (Karten et al., 2026)	arXiv	Online modify prompts, sub-agents, skills, memory; separate weight-training experiment	Pokémon Red/Emerald continuous no reset; Gemini series	Emerald Pro fixed 24 hours: improve from start 100% milestone, \$130 median vs minimal framework 98%, \$215; Flash-Lite improve reverse decline	Measured per-run API cost; main behavior cost is button-press, not model call; improver cost not separately reported	●●●
Red Queen Gödel Machine / RQGM (Iacob et al., 2026)	arXiv	Execution and evaluator code co-modify; search-score shell fixed	GPT-5.5; code, paper, proof	Polyglot held-out 71.7% vs HGM-H 69.9%, 119 vs 116 questions; each domain each baseline once	Baseline-score search tokens 1.35x-1.72x fewer; deployment per-question cost and time not measured	○●●
Group-Evolving Agents / GEA (Weng et al., 2026)	arXiv	Multiple parent agents share trajectories then modify framework code	SWE-bench/Polyglot; Haiku4.5 then Sonnet4.5, o1 reflection	Report SWE 71.0%, Polyglot 88.3%, but latter inconsistent with claimed 50-question held-out denominator; not use as generalization-gain evidence	Full cycle estimate USD 13,000 / USD 1,500; no per-question post-evolution cost change	○●●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

A05 · Whole harness (3/4)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
Hyperagents / DGM-H (Zhang et al., 2026b)	arXiv	Executor and improver shared Python program	Code, paper review, math grading; robot reward design has separate PPO training	Transfer improver to math grading imp@50=0.630, initial 0.0; 200 iterations 0.640 vs 0.610 not significant	Self-modify ~33M tokens, paper review + robot eval 0.506M/iteration; total and breakdown not fully compatible	○ ● ●
SICA (Robeyns et al., 2025)	ICLR 2025 Self-Improving Foundation Models Without Human Supervision workshop	Self-complete Python code, tools and oversight mechanism	Sonnet3.5(v2), o3-mini; fixed coding subset and synthetic-file task	SWE peak 17% → 53%, final accuracy 0.51; same baseline for agent selection, not held-out	Cross-four-benchmark full-attempt per-question average: final \$1.91 → \$1.70, 130.2 → 114.5 s; tokens 0.24 → 0.30 M; full cycle ~\$7,000	● ● ○
Harness-R1 (Shao et al., 2026)	arXiv	Frozen target-outside execution hooks; separate engineering model train SFT+GRPO	Qwen3.5-9B; WebShop, ALFWorld, DBBench	Main table same-batch 44.3% → 53.6%; also generate from 10 failures, 1,270 held-out +8.9 +/- 1.5 pp	GPU config and training data number, actual time/tokens/dollars not measured	○ ● ●
AgentDevel (Zhang, 2026)	arXiv	Prompts, code, tool wrappers one release candidate	Sonnet4.5/Claude Code; SWE, WebArena, StableToolBench	SWE Lite 11.0% → 22.0%, Verified 15.0% → 30.0%; protocol keeps test, table naming not clear	StableToolBench 11 rounds only; time/tokens/dollars not measured	○ ● ●
Darwin Gödel Machine / DGM (Zhang et al., 2026a)	ICLR 2026 main conference	Frozen model outside Python tools and workflow; search process fixed	Sonnet3.5(New) modify, SWE same model, Polyglot o3-mini execute	SWE subset 20.0% → 50.0%; Polyglot full 14.2% → 30.7%, full includes search items	80 iterations, single SWE run ~2 weeks; post-improvement per-question cost/latency not measured	○ ● ●
Live-SWE-agent (Xia et al., 2025)	arXiv	Per-question script create; framework unchanged, discard tool after completion	SWE-bench Verified/Pro; multiple backbone	Gemini3Pro success 74.2% → 77.4%; 50-question GPT-5-Nano 44.0% → 14.0%	Same Gemini average per-question cost \$0.46 → \$0.48; no offline evolution does not mean total cost down	● ● ●
RSEA (Nguyen et al., 2026a)	arXiv	Policy, skill rules, step manual text; ReAct loop unchanged	Qwen2.5-7B ALFWorld; Qwen3-30B WebShop; independent test	ALFWorld 69.3% vs ReAct 64.6%, but below Dynamic Cheatsheet 70.7 and difference not significant	Tau-bench 190 meta+selection calls total; call scope not clear; time/tokens not measured	○ ● ●
HarnessFix (Chen et al., 2026)	arXiv	Prompts, tool description, orchestration, config, adapter and validation script	GAIA, SWE-bench Verified, AppWorld, Terminal-Bench 2.0 Verified; GPT-5 mini	Initial system to patched completion +6.3 to +18.4 percentage points; mean of three runs	AppWorld offline patch 37.2 million tokens; execution-phase time/cost not reported	○ ● ●
StarHarness (Esakkiraja et al., 2026)	arXiv	Stirrup prompts, tools, skills, context, validation and end logic	ITBench / EnterpriseOps-Gym / AutomationBench, GPT-5.4	Held-out increment +31.7 / +15.1 / +29.3 pp	All-benchmark per-task inference cost estimate -17% / -53% / -29%; offline optimize cost not reported	● ○ ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

A05 · Whole harness (4/4)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
Growing Harness (L. Li et al., 2026)	arXiv	Frozen model and tools, only modify shared executable code	BrowseComp-Plus, WebArena-Verified × gpt-oss-120b/20b, Qwen3.5-4B; 50 final task per setting, 3 runs	Five of six groups highest success, one group -0.7 pp; deployment model calls -76.0-91.8%	Deployment inference cost -74.4-98.6%, not evaluator/offline improver; no break-even	● ○ ●
Sol-Pi (H. Liu et al., 2026)	arXiv	Model fixed, modify run-framework context/tool mechanisms	EdgeBench, GPT-5.6 Sol; held-out count statement inconsistent	Full config score 44.8 → 42.0, token stream -49.0%; Performance config 44.8 → 47.2, -6.1%	Measure input/cache/output token stream; not total-search tokens, dollars or time	● ○ ●
Agent JIT Compilation (Winston et al., 2026)	ICML 2026	Reuse code plan on tools, execution dispatch	Five app, GPT-4.1/Gemini-2.5-Flash/Gemini-2.5-Pro average	Planner average 11.7 s vs Browser-Use 122.1 s; 10.4x speedup	End-to-end time includes planning; plan score only dimensionless agent rank; offline total cost location entry not reported	● ○ ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

A06 · Predicting what an action does

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
WMA (Chae et al., 2025)	ICLR 2025	Train page-change prediction and scoring model; task execution model frozen	GPT-4o WebArena same-environment synthetic instruction learn; also Mind2Web train/test	WebArena 16.6% vs no-action-selection 12.8%, still below tree-search 19.2%; Mind2Web cross-task AWM+WMA 25.4% vs AWM w/o EF 15.3%	Per-instruction 140.3 s vs search 748.3 s; table API \$0.4 vs \$2.7; learn-collection cost not reported; training ~3 GPU hours total scope unclear	●●●
Curious Causality-Seeking Agents / MCG (Zhao et al., 2025)	NeurIPS 2025	Train context encoder, causal graph and change-prediction network; no language model	Chemical unseen state, Magnetic manipulator arm; CEM planning	Fork n=2 predict 63.18 ± 13.94 vs FCDL 57.82 ± 9.90; full model not every ablation winner	Only training rounds and param size tested; no time/GPU hours/intervention cost	○●●
WorldEvolver (X. Zhang et al., 2026)	arXiv	All models frozen, only modify environment model context: real change record, semantic rule, predict filter	Word2World predict, AgentBoard plan; memory cross-attempt and cross-task persist, Gemma-4-26B-A4B	ALFWorld predict exact-match 52.88 vs 3.60; plan best-of-5 26.12% vs 23.88%, first-try 19.4 below 22.4	Predict eval average 1.48 s/change vs zero-shot 1.05 s; total 5.24 vs 3.72 GPU hours; not plan-task cost	●●●
Metis (Dai et al., 2026)	arXiv	Environment facts, pitfalls, plans and callable code from repeated plans	AppWorld; offline train once, freeze memory then held-out test	GPT-4o official test TGC 60.1 vs no memory 51.8; resampled test 66.1 vs 54.8	Official per-question execution 97.4K vs 112.6K tokens, 11.25 vs 14.55 turns; does not include Memory Manager; build 7.9M tokens without trajectory collection	●○●
Meta-TTL (Lou et al., 2026)		Cross-round action prompts for same task and 'how to change prompt' prompt learned offline	Jericho, WebArena-Lite, tau2-bench; new instances and new domains	Gemini-3-Flash action, GPT-5 adaptation; WebArena-Lite unseen domain W-AUC 0.766 vs TextGrad 0.728, 3 runs, not single success rate	Jericho per session 860.9 s vs Static 593.2 s; 204.0 vs 231.0 calls; 364k vs 196k tokens; offline ~\$30-70/benchmark	●○●
AppWorld (Trivedi et al., 2024)	ACL 2024 main conference	Build resettable apps and tasks, state checker; no agent improvement	API programming operation simulate app; Test-N and Test-C include unseen app	GPT4O ReAct TGC 48.8% / 30.2%; fixed agent benchmark score, not learning gain	Current-price estimate ReAct Test-N \$0.7/task; judge average under 0.6 seconds, only checker	●—●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

A07 · Descriptions, facts and preconditions (1/2)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
Metis (Dai et al., 2026)	arXiv	Environment facts, pitfalls, plans and callable code from repeated plans	AppWorld; offline train once, freeze memory then held-out test	GPT-4o official test TGC 60.1 vs no memory 51.8; resampled test 66.1 vs 54.8	Official per-question execution 97.4K vs 112.6K tokens, 11.25 vs 14.55 turns; does not include Memory Manager; build 7.9M tokens without trajectory collection	● ○ ●
CTIM-Rover (Lindenbauer et al., 2025)	ACL 2025 REALM workshop	General and repo-level text suggestions, can include successful trajectory examples	SWE-bench Verified, 401 train, 45 test, also Lite trajectories	GPT-4o o1 extraction: full 40% vs AutoCodeRover 42%; text memory only 31%; whether extra Lite trajectories overlap not clarified	Average turns 11.16 → 9.62; final context 11414.02 → 17544.93 tokens, not cumulative tokens; time and cost not measured	● ○ ●
Meta-TTL (Lou et al., 2026)		Cross-round action prompts for same task and 'how to change prompt' prompt learned offline	Jericho, WebArena-Lite, tau2-bench; new instances and new domains	Gemini-3-Flash action, GPT-5 adaptation; WebArena-Lite unseen domain W-AUC 0.766 vs TextGrad 0.728, 3 runs, not single success rate	Jericho per session 860.9 s vs Static 593.2 s; 204.0 vs 231.0 calls; 364k vs 196k tokens; offline ~\$30-70/benchmark	● ○ ●
Agent S (Agashe et al., 2025)	ICLR 2025	Frozen model, update full-task and sub-task text memory; interface layer fixed	GPT-4o/Claude-3.5 OSWorld/WindowsAgentArena; ongoing evaluation update after exploration	GPT-4o OSWorld 20.58% vs literature baseline 11.21%; ablation subset full 26.15% vs no pre-exploration 15.38%, no independent frozen-memory held-out	Checked entry does not report steps, time, tokens, cost or build cost	○ ○ ●
WebCoach (Liu et al., 2025)	ICLR 2026 MemAgents workshop	External trajectory summary, additional Coach injects suggestions to action model	WebVoyager 643 real-time web tasks; dynamic library accumulates same-stream same-task but excludes same-task-ID retrieval	Skywork-38B + Qwen3-8B Coach: 0.614 vs 0.473; Qwen-VL-7B main table 0.311 vs 0.328	Skywork average per question 215 → 395 s, actions 10.7 → 10.2; time includes extra inference; no token cost measurement	● ○ ●
MetaAgent (Qian et al., 2025)	arXiv	Frozen model, accumulate text experience and raw tool result knowledge base; no explicit environment-rules model	QwQ-32B GAIA/WebWalkerQA/BrowseComp; pre-fill library with same-batch task simulation	WebWalkerQA 52.1 vs WebThinker-32B-RL 46.5; GAIA 47.6 below its 48.5; no held-out split	Checked entry does not report measured cost or time	○ ○ ○
GenericAgent (Liang et al., 2026)	arXiv	SOP, script, hierarchical memory; fixed tool interface	Multiple models; local task, web, repeated download and research tasks	Same GPT-5.4 repeated download 102 seconds → ~66 seconds; not held-out; five long tasks success 100%, vs Claude Code 100%, OpenClaw 80%	Repeated download tokens 200,439 → 100,000; five long tasks 220.8 s vs OpenClaw 183.1 s; evolution cost not separated	● ● ●
DRAFT / From Exploration to Mastery (Qu et al., 2025)	ICLR 2025	Rewrite natural-language tool documentation, do not train execution model	GPT-4o writes docs, GPT-4o/mini/Llama-3-70B use; RestBench, ToolBench	GPT-4o TMDb correct-path rate 88.00 vs ReAct 71.00; tool path includes standard subsequence, not final-state check	Checked entry does not report run and improvement call total, tokens, time, cost	○ ○ ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

A07 · Descriptions, facts and preconditions (2/2)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
AutoDroid-V2 (Wen et al., 2025)	ACM MobiSys 2025	App documentation, synthetic script and local model weight	DroidTask 158 task/13 app; Llama3.1 8B fine-tune; per-author explore and synthesis not look test set	54.4% vs same fine-tune step-by-step AutoDroid 43.9%	Phone model inference 46.3 s vs 669.2 s; does not include UI execution; offline prep and fine-tune separate	● ● ●
EchoPath (Zhao et al., 2026)	arXiv	Parameterized GUI trajectory memory, precondition and lifecycle	OSWorld-Verified 159 task with established memory; same-task second try change resolution, codex-gpt-5.5-medium	91.2% vs Synapse 91.8%	Median tokens 20,370 vs 586,386; time 127.5 s vs 315.7 s; first build median ~572k tokens/4.5 min counted success-only	● ● ○
Unbrowse / Shadow APIs (Tham et al., 2026)	arXiv	Shared interface graph, parameter structure, response format, auth description	94 domain read, single machine, hot cache third pass, include only aligned-output fields	Average request time 950 ms vs Playwright 3,404 ms	Measure end-to-end request time; 20-new-domain cold discover median 8,200 ms, publish success 18/20; no task cost	● ● ●
SKILL.nb (Hattami et al., 2026)	ICML 2026 FAGEN workshop	Text step, code, pre/post check and version library, frozen model	WebArena-Verified 812 task, gpt-5.3-codex, shared framework different reasoning intensity	Single round 53.7%, +3.9 pp next; three perturbation reruns consistency 91.7%	Internal maintain/update 36k tokens per success, execution included unclear; no dollars or time	● ● ●
Grounding Agent Memory (Suresh et al., 2026)	arXiv	Only modify post-task experience library, add environment read-only probe	CLBench 40 question, unreported structure change; gpt-5.4; five paired run	Pass 39% → 73%; executor per-question SQL query 8.8 → 4.7	Full execution \$3.38 → \$1.68; no experience-build and probe-maintain	● ○ ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

A08 · Page and place structure

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
CTIM-Rover (Lindenbauer et al., 2025)	ACL 2025 REALM workshop	General and repo-level text suggestions, can include successful trajectory examples	SWE-bench Verified, 401 train, 45 test, also Lite trajectories	GPT-4o o1 extraction: full 40% vs AutoCodeRover 42%; text memory only 31%; whether extra Lite trajectories overlap not clarified	Average turns 11.16 → 9.62; final context 11414.02 → 17544.93 tokens, not cumulative tokens; time and cost not measured	
MobileGPT (Lee et al., 2024)	ACM MobiCom 2024	Per-app page-subtask graph and action sequence	Mobile apps; cold start then warm start same task with parameter change	GPT-4-turbo; with human repair ablation warm start 98.75%; main table without human repair cannot mix	Human-repair path warm start vs same prompt Derive-only: latency -62.5%, cost -68.8%; offline explore per app 10 to 15 minutes, total \$10.78	
Continual Harness (Karten et al., 2026)	arXiv	Online modify prompts, sub-agents, skills, memory; separate weight-training experiment	Pokémon Red/Emerald continuous no reset; Gemini series	Emerald Pro fixed 24 hours: improve from start 100% milestone, \$130 median vs minimal framework 98%, \$215; Flash-Lite improve reverse decline	Measured per-run API cost; main behavior cost is button-press, not model call; improver cost not separately reported	
MrSteve (Park et al., 2025)	ICLR 2025	Save location, event, time and access graph; additionally train VPT-Nav, not all models frozen	Minecraft pixel control, sparse resource and long-horizon navigation; Steve-1 execute	Same sequence A-B-A task repeat A step count reduce; long-horizon ~70 tasks vs baselines <10, exact values in graph only	Memory query 11.24 ms vs FIFO 438.46 ms, not task time; navigator ~23 single-GPU training hours	
ActionEngine (Zhong et al., 2026)	arXiv	State-action graph, locator and descriptor, online general-purpose repair write-back	WebArena 655 tasks, Claude Opus 4.6; template warm-up then eval, patcher off	91.2%; no warm-up graph 73.1%	27 s vs 87 s; \$0.05 vs \$0.40, both per-task relative Claude Code; no explore/warm-up; break-even 39-101 tasks and missing warm-up	
AppAgentX (Jiang et al., 2025)	arXiv	Page/element relation memory and high-level action	AppAgent 50 task/9 app; GPT-4o; five per-question	With chain memory then add action: success 70.8% → 71.4%; steps 9.1 → 5.7	Per-task tokens 9.26k → 4.94k; per-step time 23 → 16 s, successful-task-only, build cost unclear	
Skim (Wong et al., 2026)	eligibility record: EuroSys 2027	URL template, answer structure and low-cost execution path	WebVoyager/WebShop, WebVoyager/AgentOccam/BrowserUse framework	Cost median 1.9x lower, latency -33.4%, reported accuracy no decrease	Per-task dollars and end-to-end time, include fail and retry; offline amortize location entry not reported	

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ◐ partly, ○ no (initial judgment, calc.; Appendix A1).

A09 · Exploring and practising (1/2)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
Agent S (Agashe et al., 2025)	ICLR 2025	Frozen model, update full-task and sub-task text memory; interface layer fixed	GPT-4o/Claude-3.5 OSWorld/WindowsAgentArena; ongoing evaluation update after exploration	GPT-4o OSWorld 20.58% vs literature baseline 11.21%; ablation subset full 26.15% vs no pre-exploration 15.38%, no independent frozen-memory held-out	Checked entry does not report steps, time, tokens, cost or build cost	○ ○ ●
SkillWeaver (Zheng et al., 2025)	arXiv	Playwright function library from web exploration, includes usage notes and page prerequisites	GPT-4o WebArena; self-practice then test benchmark tasks	Success rate 22.6 → 29.8; online human-judged subset 40.2 → 56.2, Cars 11.1 → 11.1	Per site 160 exploration iterations; checked entry does not report runtime and exploration time cost	○ ● ●
Voyager (G. Wang et al., 2024)	TMLR 2024	Executable Minecraft skills, completed and failed task tables	GPT-4; MineDojo/Mineflayer; exploration vs new worlds no task separation	New world each unseen task 3/3 success; ReAct, Reflexion, AutoGPT 0/3 within 50 prompting iterations	'15.3× faster' is prompt-iteration ratio to unlock wood tools; seconds, tokens, dollars checked entry not reported	● ○ ●
GenericAgent (Liang et al., 2026)	arXiv	SOP, script, hierarchical memory; fixed tool interface	Multiple models; local task, web, repeated download and research tasks	Same GPT-5.4 repeated download 102 seconds → ~66 seconds; not held-out; five long tasks success 100%, vs Claude Code 100%, OpenClaw 80%	Repeated download tokens 200,439 → 100,000; five long tasks 220.8 s vs OpenClaw 183.1 s; evolution cost not separated	● ● ●
OS-Copilot / FRIDAY (Z. Wu et al., 2024)	ICLR 2024 LLMAgents workshop	Frozen model, update program tool library; semantic memory is design component, experiment mainly validates tools	GAIA dev set learn, private test; self-created table tasks then SheetCopilot-20 test	GPT-4 SheetCopilot-20 Pass@1 60% vs no learning 0%; GAIA Level 3 both 6.12%	GAIA Level 1 table average time 118 vs 124 s; Level 2 221 vs 239 s; no-learning cost	● ○ ●
Planning to Explore / CovQValue (Amayuelas et al., 2026)	arXiv	Frozen model, store code-branch coverage graph and test history, change test-candidate selection	RepoExploreBench/TestGenEval Lite; each program independent empty library, no cross-task held-out	Gemini 3 Flash TestGenEval Lite N=24 execution average branch 108.9 vs Greedy 61.5; coverage increase but more test fail	Execution count equal, model calls not equal; no total time/tokens/cost	○ ○ ○
SEAgent (Sun et al., 2026)	ICML 2026 main conference	Update software guide and curriculum, co-train actor and interface judge, beyond directional bound	UI-TARS-7B OSWorld app subset; checked entry does not report held-out detail	Success 11.3% → 34.5%, latter first specialize then distill then generalist train; direct all-app train 30.6%, mean of three	1k train iterations, 8 A100 80GB; no GPU hours, total time, tokens	○ ● ●
WALT (Prabhu et al., 2026)	ICLR 2026 main conference	New action script, parameter constraint and tool description	WebArena, VisualWebArena; GPT-5 plan, GPT-5-mini execute	Average success 50.1% / 52.9%; same structure no-tool control best +30.7% relative, includes parser and verifier co-change	Steps best improve 1.4x; offline tokens, dollars, time not reported	● ● ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

A09 · Exploring and practising (2/2)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
AXIS (Lu et al., 2025)	ACL 2025 main conference	Translate UI trajectory into API-code skill	Microsoft Word, 50-task feasibility; location entry incomplete learn/test split list	Relative UFO: 84.0% vs 52.0%, 2.0 vs 3.2 step	29.9 s vs 59.5 s; backend \$0.2 vs \$0.4; full learn cost location entry not reported	● ○ ◐

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ◐ partly, ○ no (initial judgment, calc.; Appendix A1).

A10 · The works closest to ours (1/4)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
ActionEngine (Zhong et al., 2026)	arXiv	State-action graph, locator and descriptor, online general-purpose repair write-back	WebArena 655 tasks, Claude Opus 4.6; template warm-up then eval, patcher off	91.2%; no warm-up graph 73.1%	27 s vs 87 s; \$0.05 vs \$0.40, both per-task relative Claude Code; no explore/warm-up; break-even 39-101 tasks and missing warm-up	● ● ●
WALT (Prabhu et al., 2026)	ICLR 2026 main conference	New action script, parameter constraint and tool description	WebArena, VisualWebArena; GPT-5 plan, GPT-5-mini execute	Average success 50.1% / 52.9%; same structure no-tool control best +30.7% relative, includes parser and verifier co-change	Steps best improve 1.4x; offline tokens, dollars, time not reported	● ● ●
SpeedRunner (Huang et al., 2026)	arXiv	Frozen-model executable skill library, add/remove/modify functions	ScienceWorld, BabyAI, Crafter; gpt-5.4-mini; 200 online task, 30 held-out, 3 seed	BabyAI from ReAct ~67% to near-perfect; Crafter all-method average achievement unlock <30%	BabyAI usage ~one eighth; output-tokens vs dollars graph scope conflict, learning total cost not reported	● ● ●
ASI (Wang et al., 2025b)	COLM 2025 main conference	Python skill add to action library	WebArena online stream, no separate train set; claude-3.5-sonnet	40.4% vs plain Claude 32.7%; average action 5.0 vs 5.6, one skill = one action	Learn tokens, dollars, time not reported; step scope not equal to raw operations	● ○ ●
HarnessFix (Chen et al., 2026)	arXiv	Prompts, tool description, orchestration, config, adapter and validation script	GAIA, SWE-bench Verified, AppWorld, Terminal-Bench 2.0 Verified; GPT-5 mini	Initial system to patched completion +6.3 to +18.4 percentage points; mean of three runs	AppWorld offline patch 37.2 million tokens; execution-phase time/cost not reported	○ ● ●
StarHarness (Esakkiraja et al., 2026)	arXiv	Stirrup prompts, tools, skills, context, validation and end logic	ITBench / EnterpriseOps-Gym / AutomationBench, GPT-5.4	Held-out increment +31.7 / +15.1 / +29.3 pp	All-benchmark per-task inference cost estimate -17% / -53% / -29%; offline optimize cost not reported	● ○ ●
AutoDroid-V2 (Wen et al., 2025)	ACM MobiSys 2025	App documentation, synthetic script and local model weight	DroidTask 158 task/13 app; Llama3.1 8B fine-tune; per-author explore and synthesis not look test set	54.4% vs same fine-tune step-by-step AutoDroid 43.9%	Phone model inference 46.3 s vs 669.2 s; does not include UI execution; offline prep and fine-tune separate	● ● ●
AppAgentX (Jiang et al., 2025)	arXiv	Page/element relation memory and high-level action	AppAgent 50 task/9 app; GPT-4o; five per-question	With chain memory then add action: success 70.8% → 71.4%; steps 9.1 → 5.7	Per-task tokens 9.26k → 4.94k; per-step time 23 → 16 s, successful-task-only, build cost unclear	● ○ ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

A10 · The works closest to ours (2/4)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
Growing Harness (L. Li et al., 2026)	arXiv	Frozen model and tools, only modify shared executable code	BrowseComp-Plus, WebArena-Verified × gpt-oss-120b/20b, Qwen3.5-4B; 50 final task per setting, 3 runs	Five of six groups highest success, one group -0.7 pp; deployment model calls -76.0-91.8%	Deployment inference cost -74.4-98.6%, not evaluator/offline improver; no break-even	● ○ ●
TraceCompiler (Yadouni et al., 2026)	arXiv	Data-dependency executable workflow	AppWorld one Venmo intent, three instance; author hand-translates execution	Same instance tool calls 34 → 11; hold-one-instance 15 of 21 state test pass	Tool call only; compile cost and net save not tested	● ○ ●
EchoPath (Zhao et al., 2026)	arXiv	Parameterized GUI trajectory memory, precondition and lifecycle	OSWorld-Verified 159 task with established memory; same-task second try change resolution, codex-gpt-5.5-medium	91.2% vs Synapse 91.8%	Median tokens 20,370 vs 586,386; time 127.5 s vs 315.7 s; first build median ~572k tokens/4.5 min counted success-only	● ● ○
Unbrowse / Shadow APIs (Tham et al., 2026)	arXiv	Shared interface graph, parameter structure, response format, auth description	94 domain read, single machine, hot cache third pass, include only aligned-output fields	Average request time 950 ms vs Playwright 3,404 ms	Measure end-to-end request time; 20-new-domain cold discover median 8,200 ms, publish success 18/20; no task cost	● ● ●
SKILL.nb (Hattami et al., 2026)	ICML 2026 FAGEN workshop	Text step, code, pre/post check and version library, frozen model	WebArena-Verified 812 task, gpt-5.3-codex, shared framework different reasoning intensity	Single round 53.7%, +3.9 pp next; three perturbation reruns consistency 91.7%	Internal maintain/update 36k tokens per success, execution included unclear; no dollars or time	● ● ●
Grounding Agent Memory (Suresh et al., 2026)	arXiv	Only modify post-task experience library, add environment read-only probe	CLBench 40 question, unreported structure change; gpt-5.4; five paired run	Pass 39% → 73%; executor per-question SQL query 8.8 → 4.7	Full execution \$3.38 → \$1.68; no experience-build and probe-maintain	● ○ ●
Are Online Skill and Memory Modules Always Worth Their Tokens? (Hajimiri et al., 2026)	EMNLP 2026 main conference (listed in the programme)	Control AWM, ASI, ReasoningBank vs plain executor budget config	WebArena four domain, Gemini 3 Flash, three run	Vanilla-IB success 44.78%, 73.6K total tokens/question, that model four-config best simultaneously	Token only; plain 15 step+pruning, enhance 10 step, near-equal budget	● ● ●
ClawTrace / CostCraft (Yuan et al., 2026)	ACM CAIS 2026 Agent Skills workshop, non-archival	Trajectory-summary exported skill document, frozen model	SpreadsheetBench 30 held-out, OpenClaw + gpt-5.4; also migrate SkillsBench	Remove delete patch after two-type sum degrade 5 → 14; migrate post cost median not decrease	Execution cost with cache; SkillsBench median \$0.143 vs \$0.144; extract cost not reported	● ○ ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).

A10 · The works closest to ours (3/4)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
ESPO (L. Liu et al., 2026)	EMNLP 2026 Findings (listed in the programme)	Natural-language module prompt	Seven text benchmark, each 500 held-out; Claude Sonnet 4.5	Main optimize single average 74.67% vs GEPA 70.91%; prompt 1,004 vs 1,878 character	PUPA optimize ~3.4M tokens, 2h 13m; no deployment acceleration measured	○ ● ●
Sol-Pi (H. Liu et al., 2026)	arXiv	Model fixed, modify run-framework context/tool mechanisms	EdgeBench, GPT-5.6 Sol; held-out count statement inconsistent	Full config score 44.8 → 42.0, token stream -49.0%; Performance config 44.8 → 47.2, -6.1%	Measure input/cache/output token stream; not total-search tokens, dollars or time	● ○ ●
AXIS (Lu et al., 2025)	ACL 2025 main conference	Translate UI trajectory into API-code skill	Microsoft Word, 50-task feasibility; location entry incomplete learn/test split list	Relative UFO: 84.0% vs 52.0%, 2.0 vs 3.2 step	29.9 s vs 59.5 s; backend \$0.2 vs \$0.4; full learn cost location entry not reported	● ○ ●
Agent JIT Compilation (Winston et al., 2026)	ICML 2026	Reuse code plan on tools, execution dispatch	Five app, GPT-4.1/Gemini-2.5-Flash/Gemini-2.5-Pro average	Planner average 11.7 s vs Browser-Use 122.1 s; 10.4x speedup	End-to-end time includes planning; plan score only dimensionless agent rank; offline total cost location entry not reported	● ○ ●
Skim (Wong et al., 2026)	eligibility record: EuroSys 2027	URL template, answer structure and low-cost execution path	WebVoyager/WebShop, WebVoyager/AgentOccam/BrowserUse framework	Cost median 1.9x lower, latency -33.4%, reported accuracy no decrease	Per-task dollars and end-to-end time, include fail and retry; offline amortize location entry not reported	● ○ ●
Agent Workflow Memory (AWM) (Wang et al., 2025a)	ICML 2025	Infer text workflows from trajectories; action encapsulation has variants	WebArena online; Mind2Web cross-task, site, domain	GPT-4-0613 WebArena: 35.5 vs BrowserGym 23.5 success rate; compared input contains HTML, not strict same interface; Mind2Web full-task success rate 0.7—4.8	WebArena average steps 5.9 vs author rerun with accessibility-tree-only BrowserGym 7.9; step comparison not same as 23.5 success-rate row	● ● ●
ReasoningBank (Ouyang et al., 2026)	ICLR 2026	Text experience pool extracted from success and failure trajectories	WebArena, SWE-bench; online single-task stream	WebArena Gemini-2.5-flash success rate 40.5 → 48.8; Shopping judge accuracy 72.7%	WebArena average ReAct steps 9.7 → 8.3; another table total tokens 50,847.4 → 53,054.5, baseline foundation not specified; time and cost not measured	● ○ ●
MobileGPT (Lee et al., 2024)	ACM MobiCom 2024	Per-app page-subtask graph and action sequence	Mobile apps; cold start then warm start same task with parameter change	GPT-4-turbo; with human repair ablation warm start 98.75%; main table without human repair cannot mix	Human-repair path warm start vs same prompt Derive-only: latency -62.5%, cost -68.8%; offline explore per app 10 to 15 minutes, total \$10.78	● ● ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ○ partly, ○ no (initial judgment, calc.; Appendix A1).

A10 · The works closest to ours (4/4)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
SkillWeaver (Zheng et al., 2025)	arXiv	Playwright function library from web exploration, includes usage notes and page prerequisites	GPT-4o WebArena; self-practice then test benchmark tasks	Success rate 22.6 → 29.8; online human-judged subset 40.2 → 56.2, Cars 11.1 → 11.1	Per site 160 exploration iterations; checked entry does not report runtime and exploration time cost	○ ● ●
Metis (Dai et al., 2026)	arXiv	Environment facts, pitfalls, plans and callable code from repeated plans	AppWorld; offline train once, freeze memory then held-out test	GPT-4o official test TGC 60.1 vs no memory 51.8; resampled test 66.1 vs 54.8	Official per-question execution 97.4K vs 112.6K tokens, 11.25 vs 14.55 turns; does not include Memory Manager; build 7.9M tokens without trajectory collection	● ○ ●
GenericAgent (Liang et al., 2026)	arXiv	SOP, script, hierarchical memory; fixed tool interface	Multiple models; local task, web, repeated download and research tasks	Same GPT-5.4 repeated download 102 seconds → ~66 seconds; not held-out; five long tasks success 100%, vs Claude Code 100%, OpenClaw 80%	Repeated download tokens 200,439 → 100,000; five long tasks 220.8 s vs OpenClaw 183.1 s; evolution cost not separated	● ● ●
Harness-R1 (Shao et al., 2026)	arXiv	Frozen target-outside execution hooks; separate engineering model train SFT+GRPO	Qwen3.5-9B; WebShop, ALFWorld, DBBench	Main table same-batch 44.3% → 53.6%; also generate from 10 failures, 1,270 held-out +8.9 +/- 1.5 pp	GPU config and training data number, actual time/tokens/dollars not measured	○ ● ●
SEDM (Xu et al., 2025)	NeurIPS 2025 SEA workshop	Injectable memory fragments with utility weight and source	LoCoMo, FEVER, HotpotQA; independent build vs evaluation not clear	GPT-4o-mini FEVER: 66 vs G-Memory 62, no memory 57; HotpotQA 39 / 38 / 34	FEVER prompt tokens 2.47M vs G-Memory 3.62M, no memory 1.65M; replay cost and time cost not measured	● ○ ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ● partly, ○ no (initial judgment, calc.; Appendix A1).